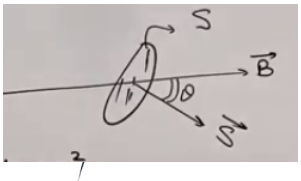
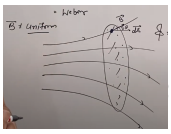
Electromagnetic Induction

Magnetic Flux: It is a physical quantity which represents the number of MLOF passing through a surface.





If $\vec{B}$ = uniform

$\phi = \vec{B}.\vec{S} = BS\cos\theta$

If $\vec{B} \neq$ uniform

$\phi = \int d\phi = \int \vec{B}.\vec{ds}$

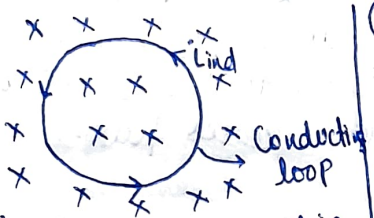
Unit = Tesla-m^2 = Weber

Faraday's law:

① In a conducting loop, if magnetic flux is changing, then Emf and current are induced in the circuit/loop.

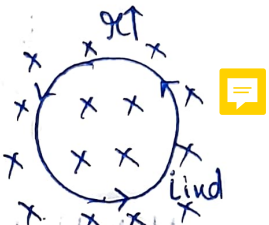
Ex

ⓐ



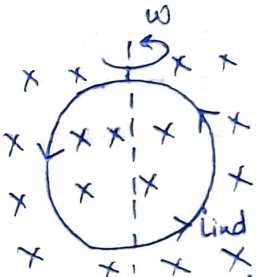
B is increased, so current is induced.

ⓑ



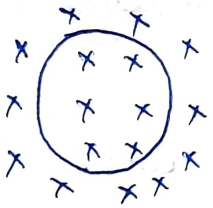
A of loop is ↑, current is induced.

ⓒ



As loop rotates, angle b/w $\vec{B}$ and $\vec{S}$ changes, so current is induced.

② Induced Emf is always equal to rate of change of magnetic flux passing through the loop.



At $t \rightarrow \phi$

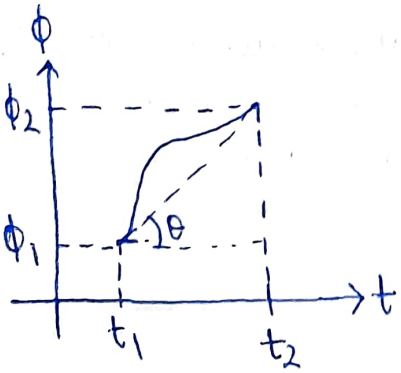
$t + \Delta t \rightarrow \phi + \Delta\phi$

$$\boxed{\mathcal{E}_{Avg} = \left|\frac{\Delta\phi}{\Delta t}\right|}$$

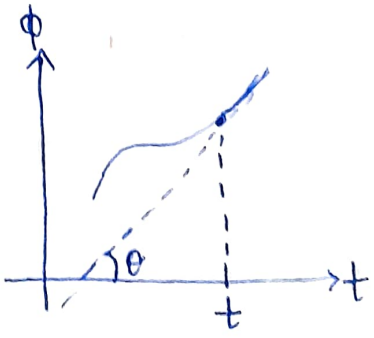
At $t \rightarrow \phi$

$t + dt \rightarrow \phi + d\phi$

$$\boxed{\mathcal{E}_{inst} = \left|\frac{d\phi}{dt}\right|}$$



$$\mathcal{E}_{Avg} = \left|\frac{\phi_2 - \phi_1}{t_2 - t_1}\right| = |\tan\theta|$$



$$\mathcal{E} = \left|\frac{d\phi}{dt}\right| = |\tan\theta|$$

Lenz Law:

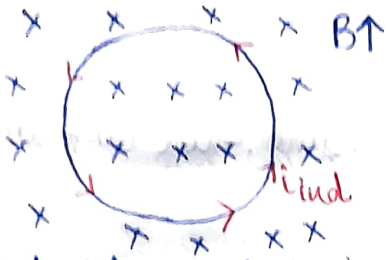
→ Induced Emf, induced current will always oppose the cause which has induced them.

→ lenz law states that induced current and induced emf will always oppose the change in magnetic flux.

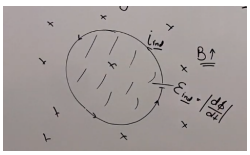
→ If B↑ → ϕ↑ → i will be induced which will try to decrease flux.
If B↓ → ϕ↓ → i will be induced which will try to increase flux.

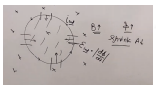
 # Lenz law states that the induced EMF and induced current will always oppose the change in magnetic flux.

Ex

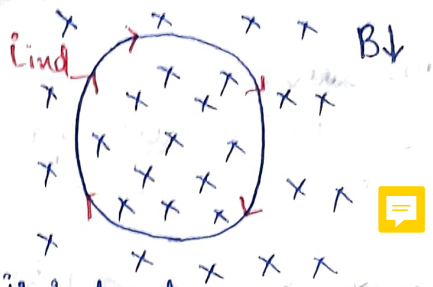


i is induced anti-clockwise so that B due to it is outwards and decreases flux.



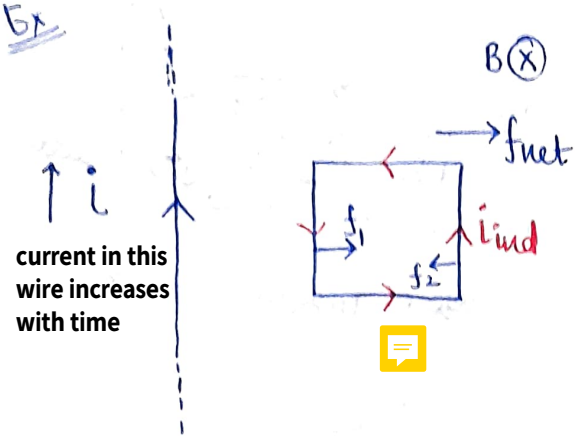


Ex



i is induced in clockwise direction so that B due to it is inwards and increases flux.

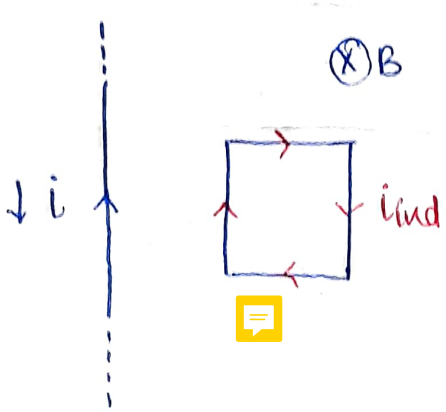
Ex



Method 1: B due to infinite wire is inside and increasing, so flux through loop↑, so current is induced anti-clockwise to produce B outwards and decrease flux.

Method 2: Fnet tries to shift the loop away from wire so as to decrease B as $B \propto \frac{1}{r}$ and tries to reduce the increasing flux.

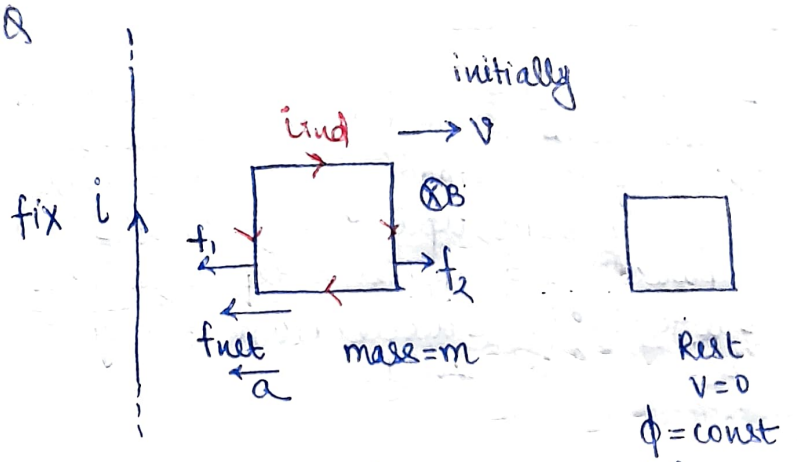
Q



Find direction of i induced.

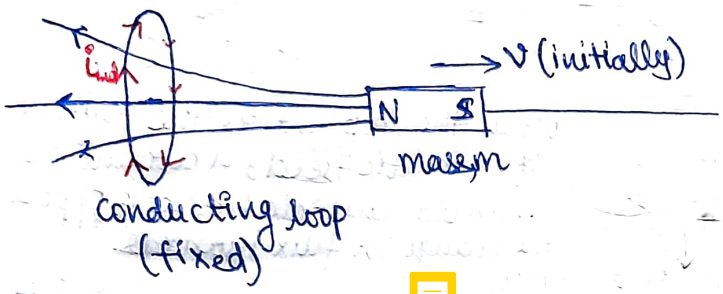
B due to infinite wire is inside and decreasing, so flux through loop↓, so current is induced clockwise to produce B inwards and increase flux.

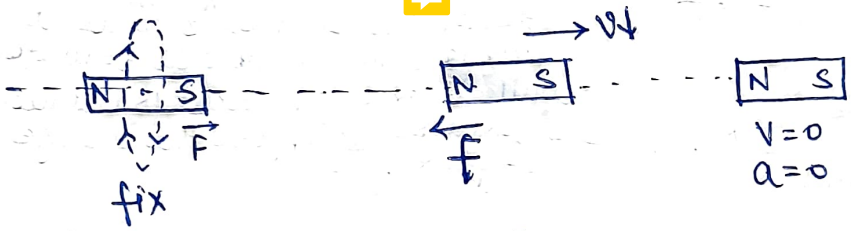
Q



$$\frac{1}{2}mv^2 = \int_0^t i^2 R\,dt$$

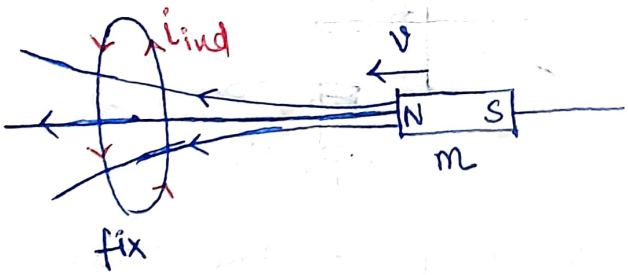
Q Gravity free space

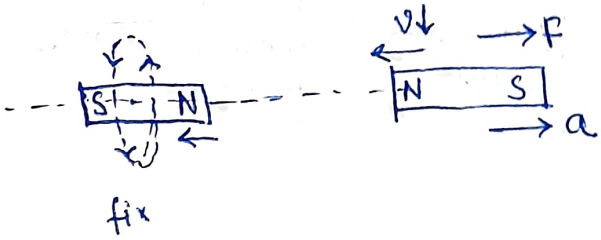




$$\frac{1}{2}mv^2 = \int i^2 R\,dt$$

Q Gravity free space





Find direction of i_{ind}

As loop moves away from wire B↓, so current is induced clockwise to ↑B. Also f_{net} acts towards wire which will reduce v. When v becomes zero and loops comes to rest, B=const ⇒ ϕ=const ⇒ i=0. Also, here mechanical energy is converted to electrical energy. KE of loop is converted and given out as heat energy

Find direction of i_{ind}.

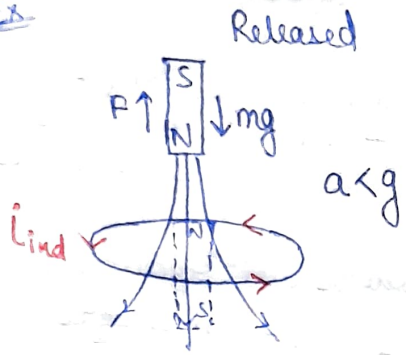
As magnet moves away flux through loop↓, so a current is induced clockwise so as to ↑ the flux in the same direcn.

Alternatively we can also say that when i is induced loop acts a magnet and attracts the other magnet. This reduces the velocity and after some time magnet will come at rest and ϕ=const. Now, no induced current flows.

Find direction of i_{ind}

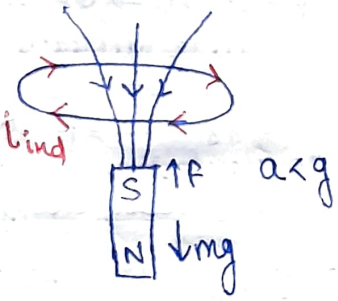
As magnet moves towards loop, flux through loop↑, so a current is induced anti-clockwise so as to reduce the flux by creating B in opposite direction.

Ex



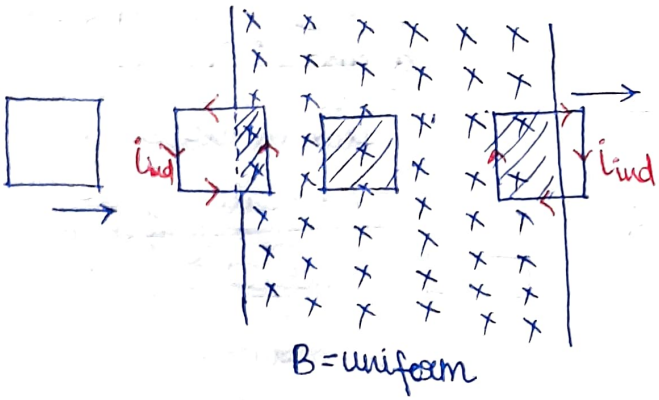
As magnet falls down flux through loop ↑, so current is induced to reduce the flux. This loop acts a magnet now with magnetic dipole moment upwards ie N pole upwards and applies a repulsive force. Due to this force acceleration of magnet becomes less than g.

Ex



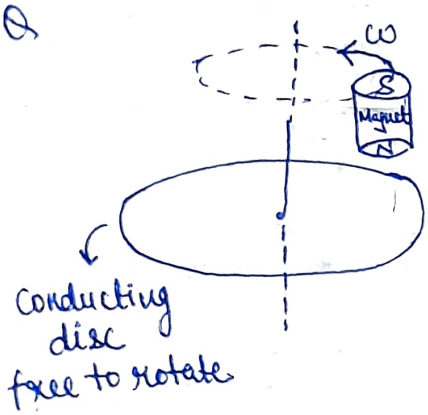
As magnet falls down flux through loop ↓, so current is induced to increase the flux in downward direcn. The loop acts a magnet now and attracts the other magnet.

Ex

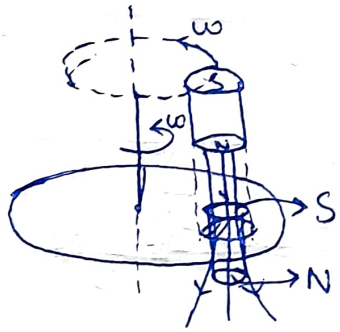


when the loop enters the region of magnetic field, a current is induced anti-clockwise to oppose the change in flux inwards. When loop completely enters the region i_{ind} becomes zero as no flux is changing now as B=uniform. While it exits a current is induced clockwise to oppose the change of flux.

*Q

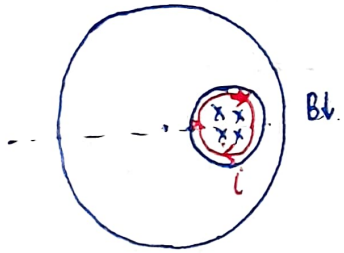


Comment on sense of rotation of disc.



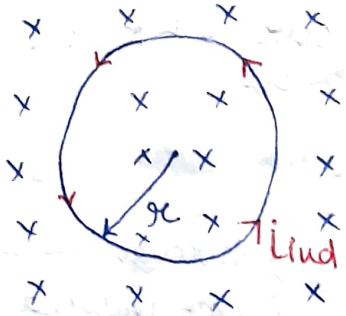
As North pole of magnet attracts the created South pole of the region of disc, Disc moves in same sense as magnet.

Top view



∴ Now, we can say that Faraday's law: $\boxed{\mathcal{E}_{ind} = -\frac{d\phi}{dt} = -\frac{\Delta\phi}{\Delta t}}$ $\mathcal{E}_{ind}$ will always oppose change in flux.

Q



Find induced emf in the loop.

$B = B_0 t$ (uniform)

n = number of turns

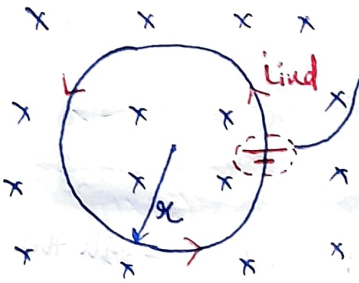
R = resistance.

$\phi = nB\pi r^2 \cos 0° = nB_0 t \pi r^2$

$\mathcal{E}_{ind} = \frac{d\phi}{dt} = nB_0 \pi r^2$

$i = \frac{\mathcal{E}}{R} = \frac{nB_0 \pi r^2}{R}$

Q



we can assume an imaginary cell of Emf = $\mathcal{E}_{ind}$ from which i_{ind} is produced.

Find induced emf in the loop.

$B = B_0 t^2$

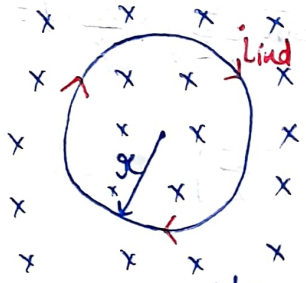
n = turns, R = resistance

$\phi = nB_0 t^2 \pi r^2 \cos 0°$

$\mathcal{E}_{ind} = \frac{d\phi}{dt} = 2nB_0 t \pi r^2$

$i = \frac{\mathcal{E}}{R} = \frac{2nB_0 t \pi r^2}{R}$

Q



Find induced emf in the loop.

$B = B_0 e^{-t/\tau}$

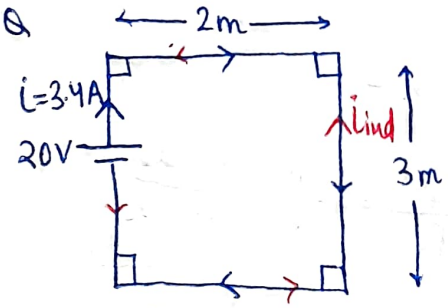
n turns, R = resistance

$\phi = nB_0 e^{-t/\tau} \pi r^2$

$\mathcal{E} = \left|\frac{d\phi}{dt}\right| = \frac{nB_0 \pi r^2 e^{-t/\tau}}{\tau}$

$i_{ind} = \frac{\mathcal{E}}{R} = \frac{nB_0 \pi r^2 e^{-t/\tau}}{\tau R}$

Q



⊗ $B = 0.5t$

$R = 5\Omega$

Find current in the circuit

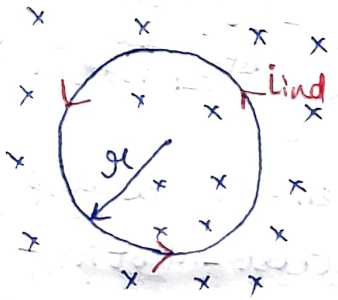
$\phi = 0.5t(6) = 3t$

$\mathcal{E}_{ind} = \frac{d\phi}{dt} = 3$ Volts

$\mathcal{E}_{eq} = 20 - 3 = 17V$

$i = \frac{17}{5} = 3.4A$

Q



B = uniform, const

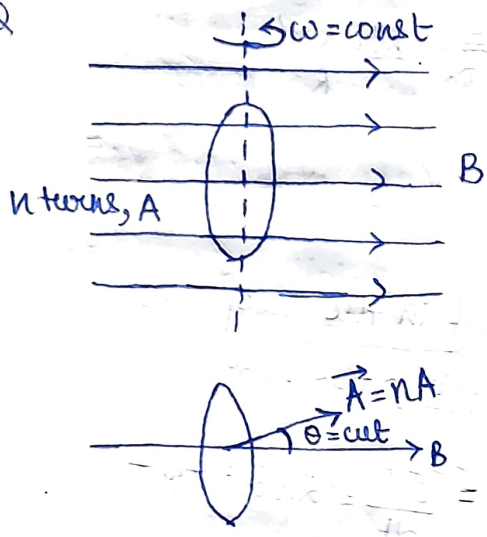
Loop expands at $\frac{dr}{dt} = k$. Find $\mathcal{E}_{ind}$

$$\phi = B\pi r^2 \cos 0^\circ = B\pi r^2$$

$$\mathcal{E} = \frac{d\phi}{dt} = B\pi 2r\frac{dr}{dt} = B\pi 2rk$$

$$\boxed{\mathcal{E} = 2\pi Bkr}$$

Q

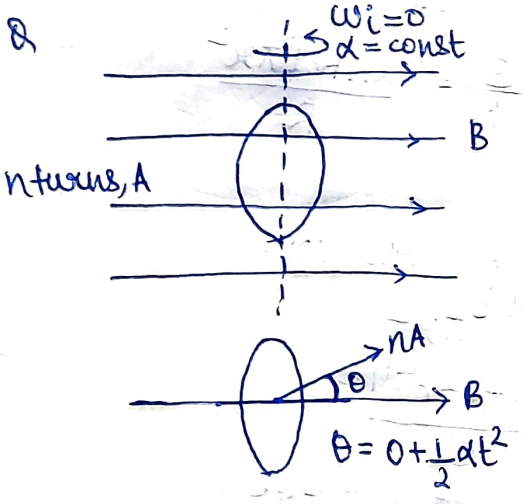


Find magnetic flux passing through the loop at any time t. Also find emf induced.

$$\phi = BnA\cos\omega t$$

$$\mathcal{E} = -\frac{d\phi}{dt} = BnA\omega\sin\omega t$$

Q



Find magnetic flux passing through the loop at any time t. Also find emf induced

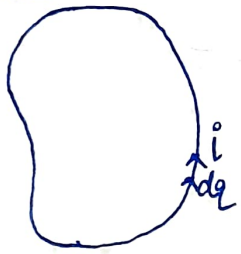
$$\phi = BnA\cos\left(\frac{\alpha t^2}{2}\right)$$

$$\mathcal{E} = -\frac{d\phi}{dt} = BnA\sin\left(\frac{\alpha t^2}{2}\right) \times \frac{\alpha}{2} \times 2t$$

$$\mathcal{E} = BnA\alpha t\sin\left(\frac{\alpha t^2}{2}\right)$$

Charge flown in circuit:

Resistance = R



$\phi = \phi(t)$

Now, $\mathcal{E} = \frac{d\phi}{dt}$

So, $i = \frac{\mathcal{E}}{R} = \frac{1}{R}\frac{d\phi}{dt}$

$\Rightarrow \int i\,dt = \frac{1}{R}\int d\phi$

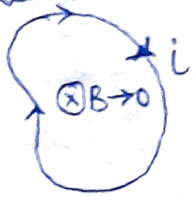
$\Rightarrow Q = \int dq = \int i\,dt = \frac{1}{R}\int_{\phi_i}^{\phi_f} d\phi$

$Q = \frac{1}{R}(\phi_f - \phi_i)$

$$\boxed{Q = \left|\frac{\Delta\phi}{R}\right| = \frac{\text{Change in flux}}{\text{Resistance}}}$$

Q Find Charge flown in circuit

(a) Resistance = R

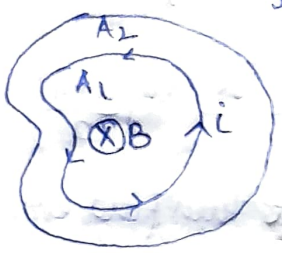


$Q = \left|\frac{\Delta\phi}{R}\right|$

$= \frac{BA-0}{R}$

$$\boxed{Q = \frac{BA}{R}}$$

(b) loop expands to area A_2

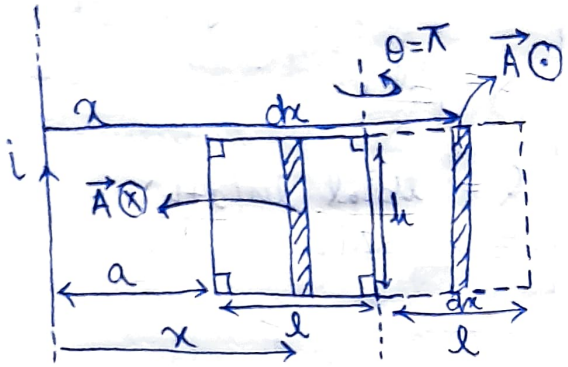


$Q = \left|\frac{\Delta\phi}{t}\right|$

$Q = \frac{BA_2 - BA_1}{R}$

$$\boxed{Q = \frac{B(A_2-A_1)}{R}}$$ anti-clockwise

Q



Resistance of loop = R

Note: If we take $\vec{A}\otimes$ in initial situation, then after rotating by π, $\vec{A}$ becomes ⊙

$$Q = \left|\frac{\Delta\phi}{R}\right| = \frac{\frac{\mu_0 i h}{2\pi}\left(\ln\left(\frac{a+l}{a}\right) + \ln\left(\frac{a+2l}{a+l}\right)\right)}{R}$$

$$\boxed{Q = \frac{\mu_0 i h}{2\pi R}\ln\left(\frac{a+2l}{a}\right)}$$

If we rotate the loop by 180°, find charge flown in loop.

$$d\phi_i = \frac{\mu_0 i}{2\pi x} h\,dx\cos 0°$$

$$\phi_i = \int d\phi_i = \frac{\mu_0 i h}{2\pi}\int_a^{a+l}\frac{dx}{x}$$

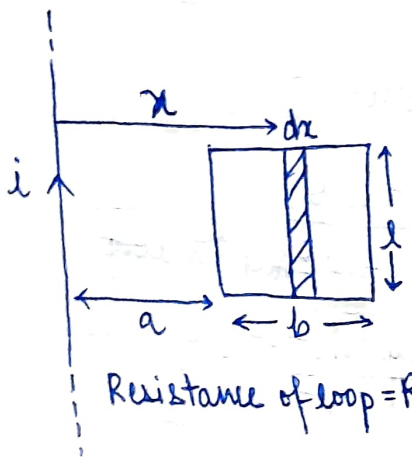
$$\boxed{\phi_i = \frac{\mu_0 i h}{2\pi}\ln\left(\frac{a+l}{a}\right)}$$

$$d\phi_f = \frac{\mu_0 i}{2\pi x} h\,dx\cos\pi$$

$$\phi_f = \int d\phi_f = -\frac{\mu_0 i h}{2\pi}\int_{a+l}^{a+2l}\frac{dx}{x}$$

$$\boxed{\phi_f = -\frac{\mu_0 i h}{2\pi}\ln\left(\frac{a+2l}{a+l}\right)}$$

Q



Resistance of loop = R

① Find magnetic flux passing through rectangular loop.

$$d\phi = \frac{\mu_0 i}{2\pi x} l\,dx\cos 0°$$

$$\phi = \int d\phi = \frac{\mu_0 i l}{2\pi}\int_a^{a+b}\frac{dx}{x}$$

$$\boxed{\phi = \frac{\mu_0 i l}{2\pi}\ln\left(\frac{a+b}{a}\right)}$$

② If $i = i_0 e^{-t/\tau}$ then find emf induced in the rectangular loop.

$$\phi = \frac{\mu_0 i_0 e^{-t/\tau} l}{2\pi} \ln\left(\frac{a+b}{a}\right) \Rightarrow \mathcal{E} = -\frac{d\phi}{dt} = \frac{\mu_0 i_0 l}{2\pi\tau} \ln\left(\frac{a+b}{a}\right) e^{-t/\tau}$$

③ If $i = i_0 \sin\omega t$ then find emf induced in the rectangular loop.

$$\phi = \frac{\mu_0 i_0 \sin\omega t\, l}{2\pi} \ln\left(\frac{a+b}{a}\right) \Rightarrow \mathcal{E} = \left|\frac{d\phi}{dt}\right| = \frac{\mu_0 i_0 \cos\omega t\, l\omega}{2\pi} \ln\left(\frac{a+b}{a}\right)$$

④ If i in wire changes from i_0 to 0 in Δt, find $\mathcal{E}_{Avg}$ in loop. Also find charge flown in loop.

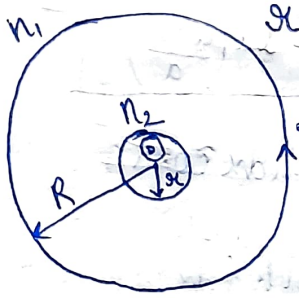
$$\phi_i = \frac{\mu_0 i_0 l}{2\pi} \ln\left(\frac{a+b}{a}\right) \qquad \phi_f = 0$$

$$\mathcal{E}_{Avg} = \left|\frac{\Delta\phi}{\Delta t}\right| = \frac{\frac{\mu_0 i_0 l}{2\pi} \ln\left(\frac{a+b}{a}\right)}{\Delta t}$$

$$Q = \left|\frac{\Delta\phi}{R}\right|$$

$$Q = \frac{\mu_0 i_0 l}{2\pi R} \ln\left(\frac{a+b}{a}\right)$$

Q



$r \ll R$

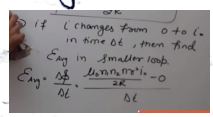
Coplanar, concentric loops.

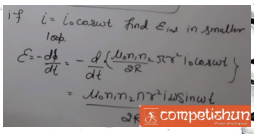
① Find magnetic flux passing through smaller loop.

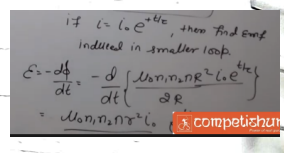
$$B = \frac{\mu_0 i n_1}{2R}$$

$$\phi = \frac{\mu_0 n_1 i}{2R} (\pi r^2) n_2 \cos 0°$$

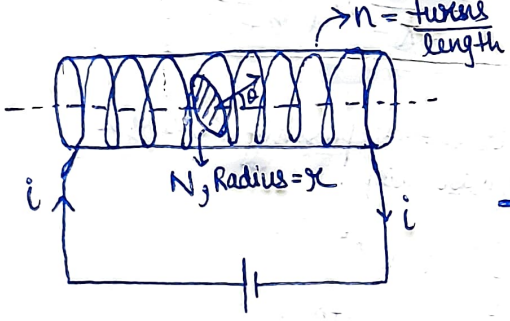
$$\boxed{\phi = \frac{\mu_0 n_1 n_2 i \pi r^2}{2R}}$$



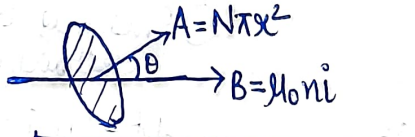




Q



① Find magnetic flux passing through small circular loop.



$$\boxed{\phi = (\mu_0 n i)(N\pi r^2)\cos\theta}$$

② If $i \to 0$ in time Δt, find $\mathcal{E}_{Avg}$ in loop.

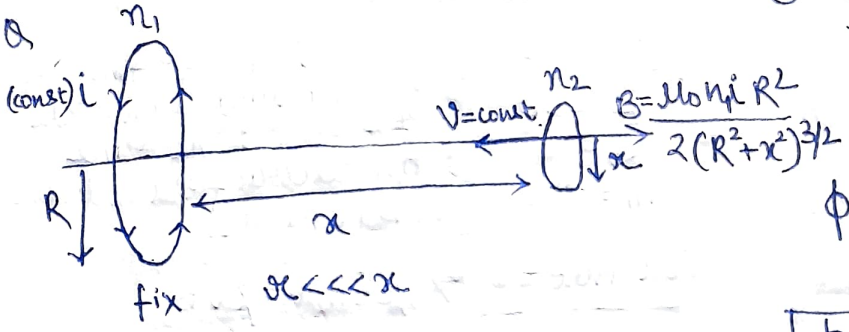
$$\boxed{\mathcal{E}_{avg} = \frac{\Delta\phi}{t} = \frac{\mu_0 n i N \pi r^2 \cos\theta}{t}}$$

③ If $i = i_0 \sin\omega t$ then find $\mathcal{E}_{ind}$ in loop.

$$\phi = (\mu_0 n i_0 \sin\omega t)(N\pi r^2)\cos\theta$$

$$\boxed{\mathcal{E} = \frac{d\phi}{dt} = \mu_0 n N \pi r^2 \cos\theta\, i_0 \omega \cos\omega t}$$

Q



① Find magnetic flux passing through small circular loop due to current in larger loop

$$\phi = \frac{\mu_0 n_1 i R^2}{2(R^2+x^2)^{3/2}} \, n_2 \pi r^2 \cos 0^\circ$$

$$\boxed{\phi = \frac{\mu_0 n_1 n_2 i R^2 \pi r^2}{2(R^2+x^2)^{3/2}}}$$

② Find emf induced in small circular loop.

$$\varepsilon = \left|\frac{d\phi}{dt}\right| = \frac{\mu_0 n_1 n_2 i R^2 \pi r^2}{2} \frac{d}{dt}\left((R^2+x^2)^{-3/2}\right)$$

$$\varepsilon = \frac{\mu_0 n_1 n_2 i R^2 \pi r^2}{2} \cdot \frac{-3}{2}(R^2+x^2)^{-5/2} \cdot \left(2x\frac{dx}{dt}\right) = \frac{\mu_0 n_1 n_2 i R^2 \pi r^2}{2} \cdot \frac{-3}{2}(R^2+x^2)^{-5/2}(2x(-v))$$

$$\boxed{\varepsilon = \frac{3\mu_0 n_1 n_2 i R^2 \pi r^2 v}{2} \cdot \frac{x}{(R^2+x^2)^{5/2}}} \qquad \left\{\frac{dx}{dt} = -v\right\}$$

③ Find the value of x for which ε_{ind} is maximum

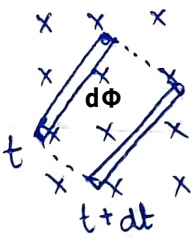
$$\frac{d\varepsilon}{dx} = \frac{3\mu_0 n_1 n_2 i R^2 \pi r^2 v}{2} \left\{\frac{(R^2+x^2)^{5/2} \cdot 1 - x \cdot \frac{5}{2}(R^2+x^2)^{3/2} \cdot 2x}{((R^2+x^2)^{5/2})^2}\right\} = 0$$

$$\rightarrow (R^2+x^2)^{5/2} = 5x^2(R^2+x^2)^{3/2} \Rightarrow R^2+x^2 = 5x^2 \Rightarrow R^2 = 4x^2$$

$$\boxed{x = \pm \frac{R}{2}}$$

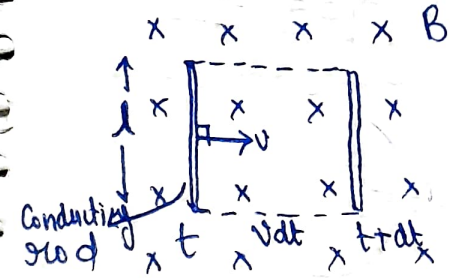
Motional Emf:

→ If a conducting wire or rod cuts magnetic lines of forces, then some emf is induced across it.



$$\boxed{\varepsilon_{ind} = \frac{d\phi}{dt} = \frac{\text{Flux cut}}{\text{Time taken}}}$$

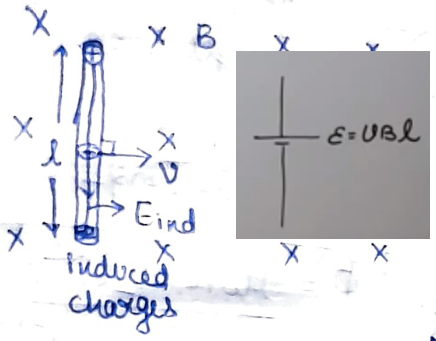
Case I: $\vec{l} \perp \vec{v} \perp \vec{B}$

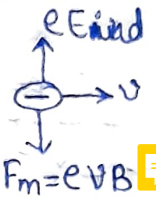


$$\varepsilon = \frac{d\phi}{dt} = \frac{Bl(vdt)}{dt} = Blv$$

$$\star \boxed{\varepsilon = vBl}$$

We can also analyze the same using Lorentz Force:





Due to Fm on $e^{\ominus}$ they start moving downwards and after some time an electric field is developed by charges.

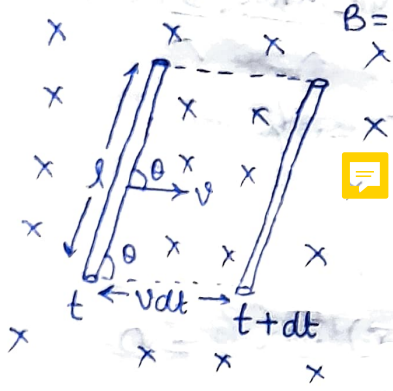
For $F_{net}=0 \rightarrow eE_{ind} = evB$

$$\boxed{E_{ind} = vB}$$

Now, $\boxed{\Delta V = El = vBl}$

(*) $\vec{v} \times \vec{B}$ will provide direction of the positive polarity.

Case 2:



$$\varepsilon = \frac{d\phi}{dt} = \frac{Blv\,dt\sin\theta}{dt} = Blv\sin\theta$$

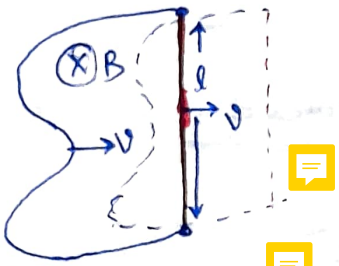
$$\boxed{\varepsilon = vBl\sin\theta}$$

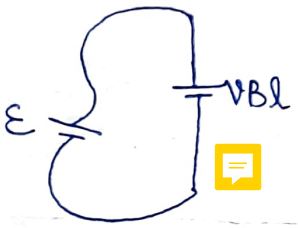
→ If $\theta = 90^\circ \Rightarrow \varepsilon = vBl$

→ If $\theta = 0^\circ \Rightarrow \varepsilon = 0$ (No MLOF is cut)

(*) If out of $\vec{B}, \vec{v}, \vec{l}$; any two vectors are parallel or antiparallel then no MLOF will be cut $\Rightarrow \varepsilon_{ind} = 0$

Case 3: For arbitrary shaped conductor.





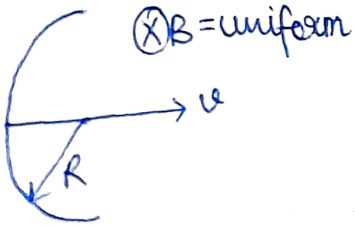
Using KVL:

$vBl - \varepsilon = 0$

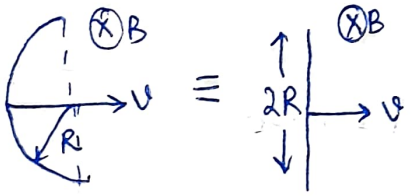
$\therefore \boxed{\varepsilon = vBl}$

(*) Arbitrary conductor can be replaced by a straight rod connecting its ends. Emf induced will be same.

Q



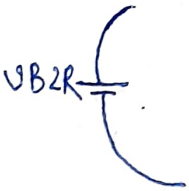
Find ε_{ind}

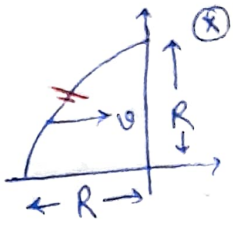


$\varepsilon = vBl$

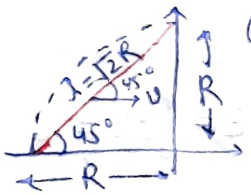
$\rightarrow \boxed{\varepsilon = vB2R}$

Also, $\vec{v} \times \vec{B}$ gives direction of polarity



Q 

Find $\mathcal{E}_{ind}$

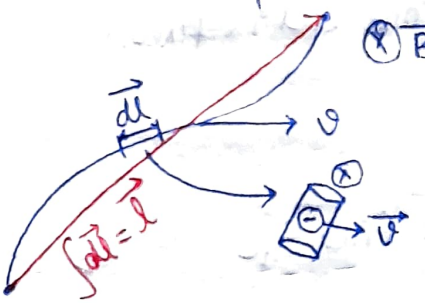


$\mathcal{E} = vBl\sin\theta$

$\mathcal{E} = vB(\sqrt{2}R)\left(\frac{1}{\sqrt{2}}\right)$

$\boxed{\mathcal{E} = vBR}$

Mathematical expression of motional emf :



$\vec{F_m} = (-e)(\vec{v}\times\vec{B})$

$\vec{F_e} = -e\vec{E}_{ind}$

Now, $\vec{F_m} + \vec{F_e} = 0$

$(\vec{v}\times\vec{B}) + \vec{E}_{ind} = 0 \Rightarrow \boxed{\vec{E}_{ind} = -(\vec{v}\times\vec{B})}$

Now, $dV = -\vec{E}\cdot d\vec{l} = (\vec{v}\times\vec{B})\cdot d\vec{l}$

$\mathcal{E} = \int dV = \int (\vec{v}\times\vec{B})\cdot d\vec{l}$

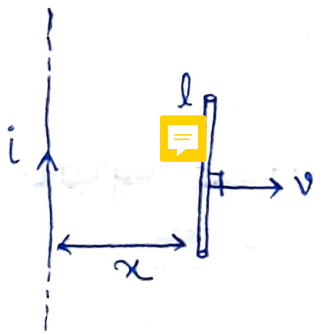
If $\vec{v}$ = same and $\vec{B}$ = same

$\mathcal{E} = (\vec{v}\times\vec{B})\cdot\int d\vec{l}$

$\star\ \boxed{\mathcal{E} = (\vec{v}\times\vec{B})\cdot\vec{l}}$

= scalar triple product
= box product

⊛ If magnetic field is uniform and velocity of each point of conducting wire is same, then we can replace the wire by a straight rod from initial to final point to find $\mathcal{E}_{ind}$.

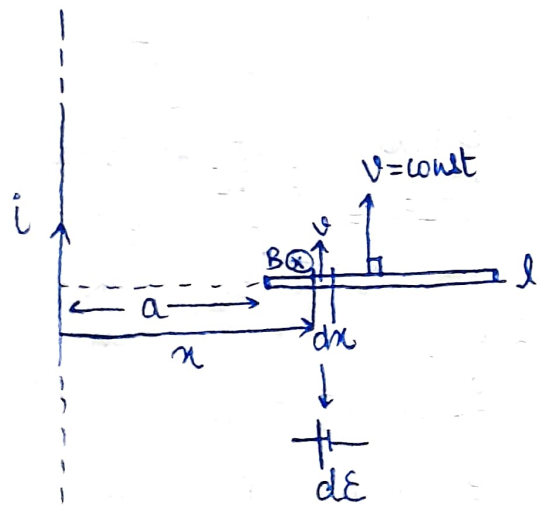
Q 

Find Emf induced in rod at this instant.

v and B at every point of rod is same.

Also, $\vec{v} \perp \vec{l} \perp \vec{B}$

$\mathcal{E} = vBl = v\left(\frac{\mu_0 i}{2\pi x}\right)l = \frac{vl\mu_0 i}{2\pi x}$

Q 

Find emf induced in rod.

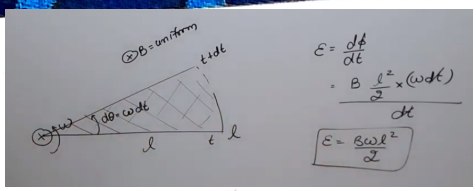
$d\mathcal{E} = v\frac{\mu_0 i}{2\pi x}dx$

All such Emf will be in series.

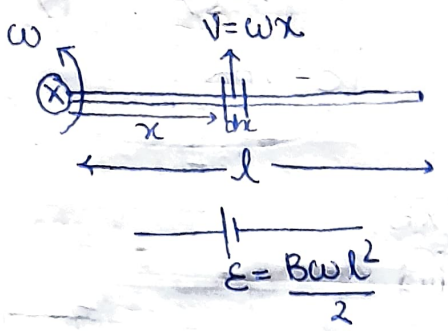
$\mathcal{E} = \int d\mathcal{E} = \frac{v\mu_0 i}{2\pi}\int_a^{a+l}\frac{dx}{x}$

$\boxed{\mathcal{E} = \frac{v\mu_0 i}{2\pi}\ln\left(\frac{a+l}{a}\right)}$

Emf induced in rotating rod:



(1) B = uniform

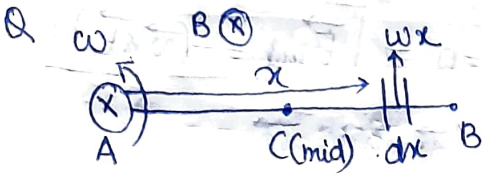


$d\varepsilon = (\omega x) B\, dx$

$\varepsilon = \int d\varepsilon = B\omega \int_0^l x\, dx = \frac{B\omega l^2}{2}$

$\therefore \boxed{\varepsilon = \frac{B\omega l^2}{2}}$ Note: This is only valid if one end is fixed.

Q



Given $\varepsilon_{AB} = 100V$. Find ε_{AC} and ε_{CB}

$\varepsilon_{AB} = \frac{B\omega l^2}{2} = 100V$ — (1)

(i) $\varepsilon_{AC} = \frac{B\omega (l/2)^2}{2} = \frac{B\omega l^2}{8} = \frac{1}{4}(\varepsilon_{AB}) = 25V$

(ii) ε_{CB} {As neither C nor B is fixed, we cannot use formula}

Method 1:

$\varepsilon_{CB} + \varepsilon_{AC} = \varepsilon_{AB}$

$\varepsilon_{CB} = 100 - 25 = 75V$

Method 2:

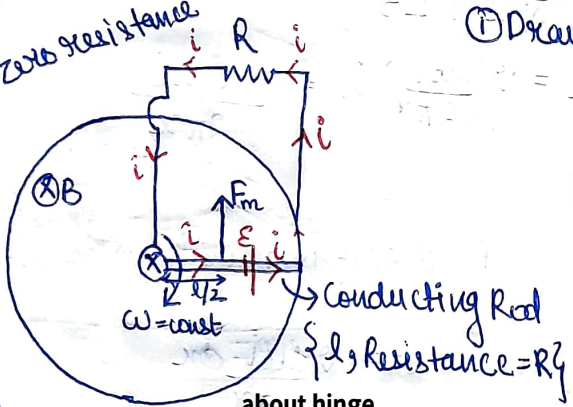
$\varepsilon_{CB} = \omega B \int_{l/2}^{l} x\, dx = \frac{3B\omega l^2}{8} = \frac{3}{4}\left(\frac{B\omega l^2}{2}\right)$

$\varepsilon_{CB} = \frac{3}{4} \times 100 = 75V$.

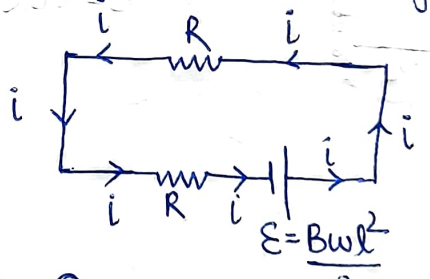
Q

Top view

Loop has zero resistance



(1) Draw equivalent circuit diagram.



(2) Find current through rod

$i = \frac{\varepsilon}{R_{eq}} = \frac{\frac{B\omega l^2}{2}}{2R} = \frac{B\omega l^2}{4R}$

(3) Find magnetic force on rod.

$\vec{F}_m = i(\vec{l} \times \vec{B}) = ilB\sin 90° = ilB$

$F_m = \frac{B\omega l^2}{4R} \cdot l \cdot B$

$\boxed{F_m = \frac{B^2\omega l^3}{4R}}$

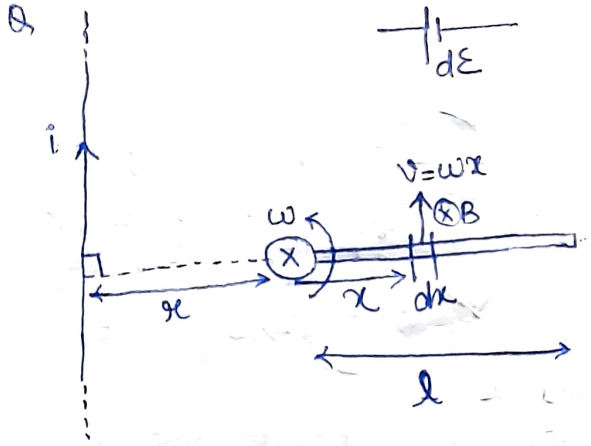
(4) Find external torque to maintain const ω.

F_m will try to reduce ω, but ω is const, so there must be an external force which will provide equal and opp. torque.

$\tau_{ext} = \tau_{Fm} = F\, l/2$

$\boxed{\tau_{ext} = \frac{B^2\omega l^4}{8R}}$ {Clockwise}

Q



Find Emf induced in rod at this instant.

$$d\varepsilon = (\omega x)\left(\frac{\mu_0 i}{2\pi(x+x)}\right)dx$$

$$\varepsilon = \int d\varepsilon = \frac{\omega\mu_0 i}{2\pi}\int_0^l \frac{x\,dx}{x+x}$$

$$\varepsilon = \frac{\omega\mu_0 i}{2\pi}\left\{x - x\ln(x+x)\right\}_0^l$$

$$I = \int \frac{x\,dx}{x+x} = \int \frac{(x+x-x)dx}{x+x}$$

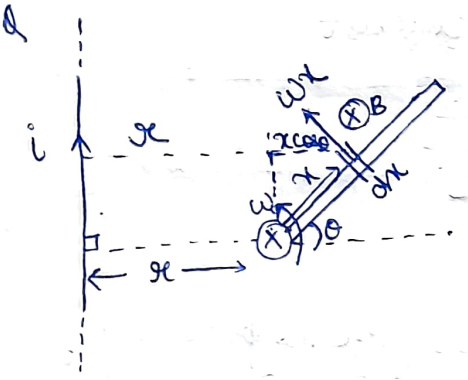
$$= \int dx - x\int \frac{dx}{x+x}$$

$$= x - x\ln(x+x)$$

$$\varepsilon = \frac{\omega\mu_0 i}{2\pi}\left\{l - x\ln(x+l) + x\ln x\right\}$$

$$\boxed{\varepsilon = \frac{\omega\mu_0 i}{2\pi}\left\{l + x\ln\frac{x}{x+l}\right\}}$$

Q



Find Emf induced in rod at this instant.

$$d\varepsilon = (\omega x)\left(\frac{\mu_0 i}{2\pi(x+x\cos\theta)}\right)dx$$

$$\varepsilon = \int d\varepsilon = \frac{\omega\mu_0 i}{2\pi}\int_0^l \frac{x\,dx}{x+x\cos\theta}$$

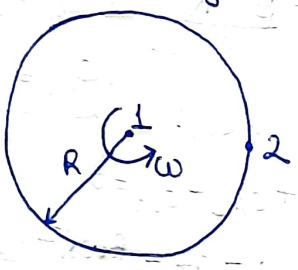
Solve...

Q***

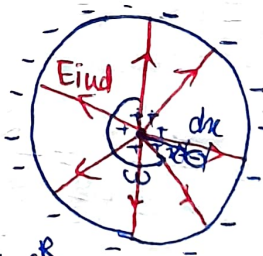
Given mass of $e^{\ominus} = m_e$ and charge $= -e$

Find emf induced b/w point ① and ②.

Conducting disc



As the disc rotates, $e^{\ominus}$ experiences a centrifugal force due to which they start moving towards surface and at centre we have deficiency of $e^{\ominus}$



$$eE_{ind} = m_e\omega^2 x$$

$$\boxed{E_{ind} = \frac{m_e\omega^2}{e}x}$$

$$dV = E_{ind}\,dx$$

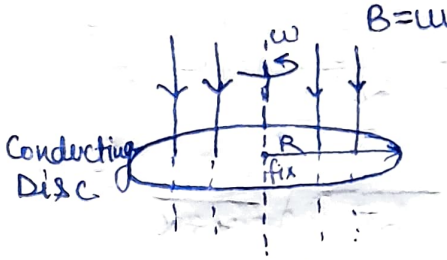
$$dV = \frac{m_e\omega^2}{e}x\,dx$$

$$\varepsilon = \int dV = \frac{m_e\omega^2}{e}\int_0^R x\,dx = \frac{m_e\omega^2}{e}\frac{R^2}{2}$$

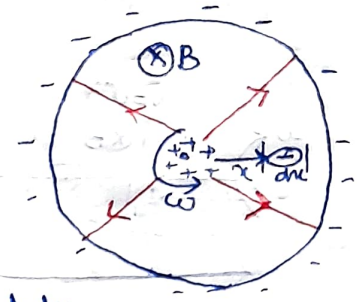
$$\therefore \boxed{\varepsilon = \frac{m_e\omega^2 R^2}{2e}}$$

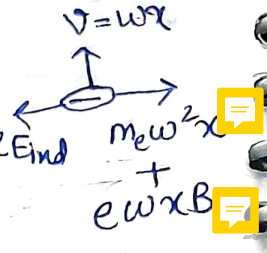
This value is negligible as m_e in order of 10^{-31} and e is in order of 10^{-19}, so ε comes in order of 10^{-12}.

Emf induced in a conducting disc :



Top view





$eE_{ind} = m_e\omega^2 x + e\omega x B$

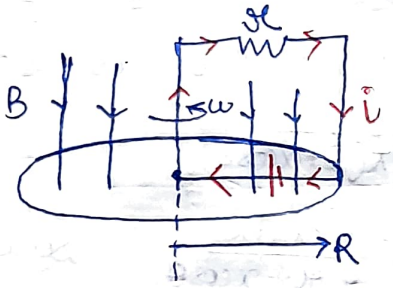
$$\boxed{E_{ind} = \frac{m_e\omega^2 x}{e} + \omega B x}$$

$dV = E_{ind}\,dx$

$$\mathcal{E} = \int dV = \int_0^R \left(\frac{m_e\omega^2 x}{e} + \omega B x\right) dx$$

$$\mathcal{E} = \frac{m_e\omega^2 R^2}{2e} + \frac{\omega B R^2}{2} \Rightarrow \boxed{\mathcal{E} = \frac{B\omega R^2}{2}}$$

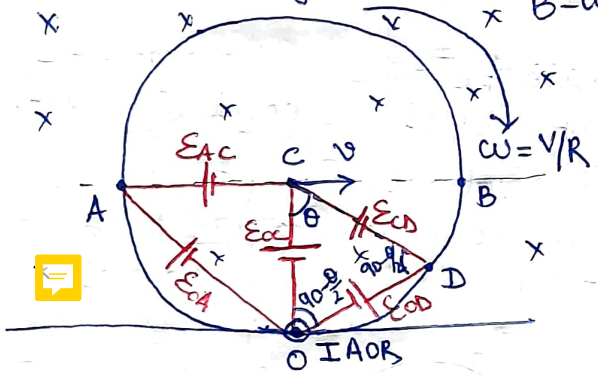
Q



Find current in circuit

$$i = \frac{\mathcal{E}}{R} = \frac{B\omega R^2}{2r}$$

Q**



Disc performs pure rolling.

Find ① $\mathcal{E}_{OC}$ ② $\mathcal{E}_{OA}$ ③ $\mathcal{E}_{AC}$ ④ $\mathcal{E}_{OD}$ ⑤ $\mathcal{E}_{CD}$

① $\boxed{\mathcal{E}_{OC} = \frac{B\omega R^2}{2}}$

② $\mathcal{E}_{OA} = \frac{B\omega(2R)^2}{2}$

$\boxed{\mathcal{E}_{OA} = B\omega R^2}$

③ $\mathcal{E}_{AC}$ ⇒ Using KVL:

$\mathcal{E}_{OC} + \mathcal{E}_{AC} - \mathcal{E}_{OA} = 0$

$\mathcal{E}_{AC} = \mathcal{E}_{OA} - \mathcal{E}_{OC} = B\omega R^2 - \frac{B\omega R^2}{2}$

$\boxed{\mathcal{E}_{AC} = \frac{B\omega R^2}{2}}$

④ $\mathcal{E}_{OD}$ ⇒ Let $OD = x$

$$\frac{x}{\sin\theta} = \frac{R}{\sin(90-\theta/2)} \Rightarrow \frac{x}{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}} = \frac{R}{\cos\theta/2}$$

$\boxed{x = 2R\sin\theta/2}$

$$\mathcal{E}_{OD} = \frac{B\omega\, 4R^2\sin^2\theta/2}{2}$$

$\boxed{\mathcal{E}_{OD} = 2B\omega R^2\sin^2\theta/2}$

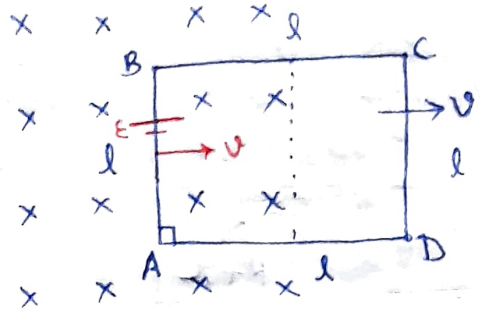
⑤ $\mathcal{E}_{CD}$ ⇒ Using KVL:

$\mathcal{E}_{OD} + \mathcal{E}_{CD} - \mathcal{E}_{OC} = 0$

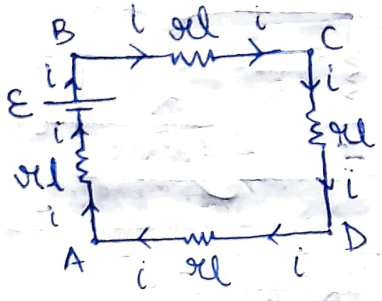
$\mathcal{E}_{CD} = \mathcal{E}_{OC} - \mathcal{E}_{OD}$

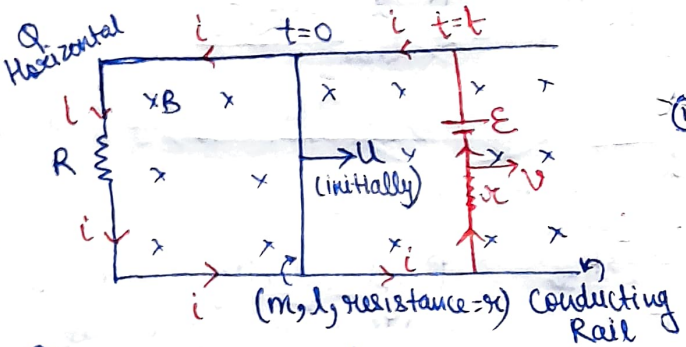
$$\boxed{\mathcal{E}_{CD} = \frac{B\omega R^2}{2} - 2B\omega R^2\sin^2\theta/2}$$

Q

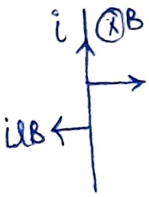


B = uniform
Given, Resistance pu. length = r





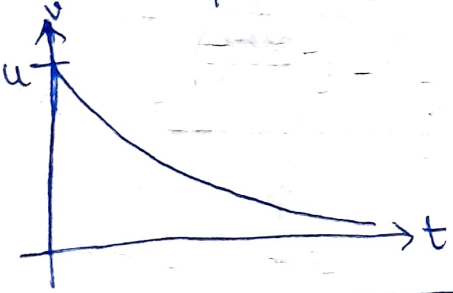
② Apply NL on rod,



$$-ilB = m\frac{dv}{dt} \quad -②$$

{-ve indicates ilB is reducing v}

④ Draw v v/s t



⑥ Find P_R

$$P_R = i^2R = \frac{u^2B^2l^2}{(R+r)^2} e^{\frac{-2B^2l^2t}{m(R+r)}} R$$

Find ① ε_{ind} ⇒ Emf is induced only in AB as CD is out of B magnetic field and for BC and AD, $\vec{B} \parallel \vec{v}$.

$$\boxed{\varepsilon = vBl}$$

② Find current:

$$i = \frac{\varepsilon}{R_{eq}} = \frac{vBl}{4rl} = \frac{vB}{4r}$$

③ Find (i) $V_{AB} = \varepsilon - iR = vBl - \frac{vBl}{4rl} \cdot rl$

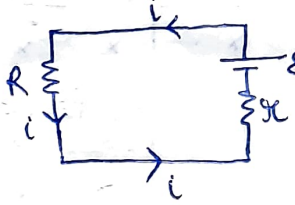
$$\boxed{V_{AB} = \frac{3vBl}{4}}$$

(ii) $V_{BC} = iR = \frac{vBl}{4rl} \times rl = \frac{vBl}{4}$

(iii) $V_{CD} = iR = \frac{vBl}{4rl} \times rl = \frac{vBl}{4} = V_{AB}$

At any time t,

① Draw equivalent circuit diag.



$$\boxed{i = \frac{vBl}{R+r}} \quad -①$$

③ Find v and i at any time t.

$$-ilB = m\frac{dv}{dt} \Rightarrow -\frac{vB^2l^2}{R+r} = m\frac{dv}{dt} \quad \{\text{from } ①\}$$

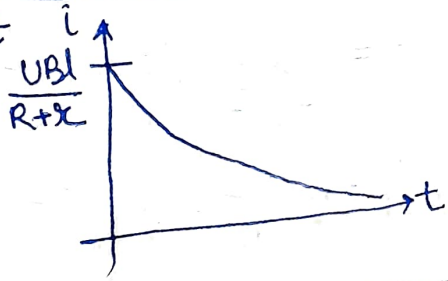
$$\Rightarrow \int_u^v \frac{dv}{v} = \frac{-B^2l^2}{m(R+r)} \int_0^t dt$$

$$= \ln\left(\frac{v}{u}\right) = \frac{-B^2l^2}{m(R+r)} t$$

$$= \boxed{v = ue^{\frac{-B^2l^2t}{m(R+r)}}}$$

$$\boxed{i = \frac{vBl}{R+r} = \frac{uBl}{R+r} e^{\frac{-B^2l^2t}{m(R+r)}}}$$

⑤ Draw i v/s t



(f) Find total heat dissipated by R.

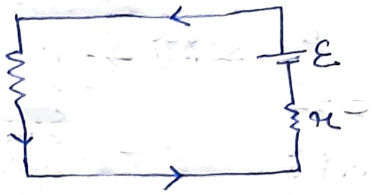
$$H_R = \int P_R\,dt = \frac{v^2B^2l^2R}{(R+x)^2}\int_0^\infty e^{\frac{-2B^2l^2t}{m(R+x)}}dt = \frac{v^2B^2l^2R}{(R+x)^2}\cdot\frac{m(R+x)}{(-2B^2l^2t)}\left\{e^{\frac{-2B^2l^2t}{m(R+x)}}\right\}_0^\infty$$

$$H_R = \frac{mv^2R}{2(R+x)}\{0-1\} = \frac{1}{2}\frac{mv^2R}{R+x} \quad \therefore \boxed{H_R = \frac{1}{2}mv^2\frac{R}{R+x}}$$

Alternate: Also, we can say that all the kinetic energy of rod gets dissipated as heat.

$$\frac{H_R}{H_x} = \frac{R}{x} \quad -(1)$$

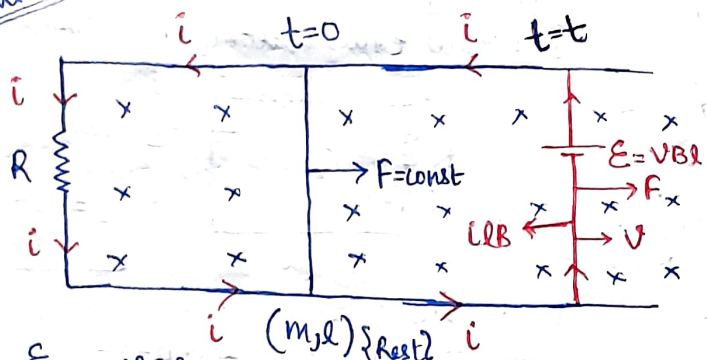
$$P_R = i^2R$$
$$P_x = i^2x$$



$$\frac{1}{2}mv^2 = H_R + H_x \quad -(2)$$

$$\frac{1}{2}mv^2 = H_R + \frac{H_R x}{R} \Rightarrow \frac{1}{2}mv^2 = H_R\left(\frac{R+x}{R}\right) \Rightarrow \boxed{H_R = \frac{1}{2}mv^2\frac{R}{R+x}}$$

Q Horizontal



Find velocity of rod and current in circuit at any time t.

$$i = \frac{\varepsilon}{R} = \frac{VBl}{R} \quad -(1)$$

$$F - ilB = m\frac{dv}{dt} \quad -(2)$$

From (1)

$$F - \frac{vB^2l^2}{R} = m\frac{dv}{dt}$$

$$\Rightarrow \int_0^t dt = m\int_0^v \frac{dv}{F - \frac{vB^2l^2}{R}}$$

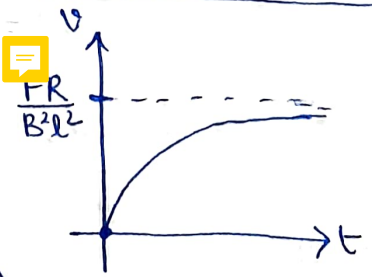
$$\Rightarrow t = \frac{mR}{-B^2l^2}\ln\left(F - \frac{vB^2l^2}{R}\right)_0^v$$

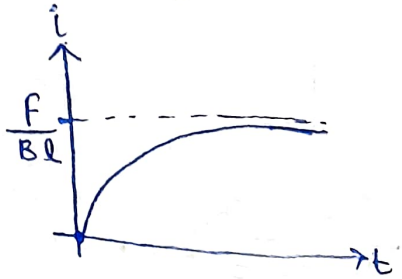
$$\Rightarrow \frac{-B^2l^2t}{mR} = \ln\left(\frac{F - \frac{vB^2l^2}{R}}{F}\right)$$

$$\Rightarrow 1 - \frac{vB^2l^2}{FR} = e^{\frac{-B^2l^2t}{mR}}$$

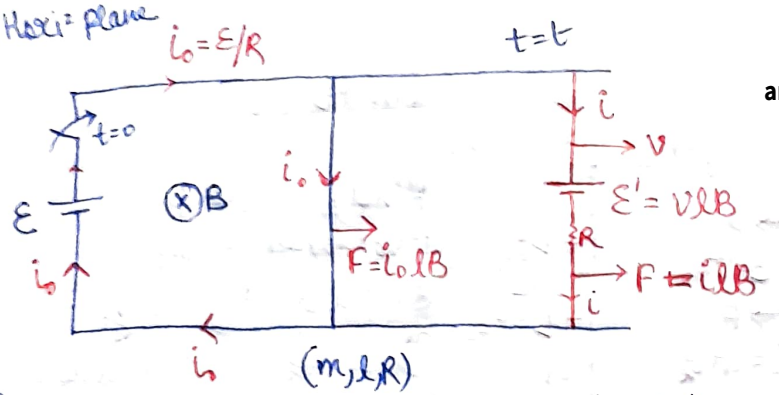
$$\Rightarrow \boxed{v = \frac{FR}{B^2l^2}\left(1 - e^{\frac{-B^2l^2t}{mR}}\right)}$$

$$\boxed{i = \frac{VBl}{R} = \frac{F}{Bl}\left(1 - e^{\frac{-B^2l^2t}{mR}}\right)}$$





Q Horizontal plane



At any time t, find velocity and current in circuit.

ans. As we close the switch current i_0 flows in circuit. As it passes through rod a force is applied on it due to which rod moves and an emf develops in it.

$$i = \frac{\mathcal{E}_{eq}}{R_{eq}} = \frac{\mathcal{E} - \mathcal{E}'}{R} = \frac{\mathcal{E} - vBl}{R} \quad -(1)$$

$$ilB = m\frac{dv}{dt} \quad -(2)$$

From (1)

$$\frac{\mathcal{E}lB - vB^2l^2}{R} = m\frac{dv}{dt}$$

$$\left(\frac{\mathcal{E} - vBl}{R}\right)lB = m\frac{dv}{dt}$$

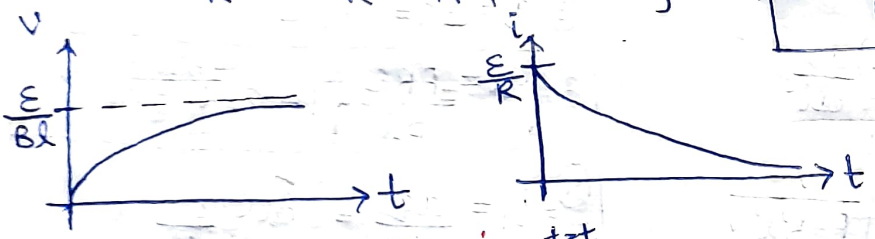
$$\Rightarrow \frac{1}{mR}\int_0^t dt = \frac{1}{lB}\int_0^v \frac{dv}{\mathcal{E} - vBl}$$

$$\frac{t}{mR} = \frac{-1}{B^2l^2}\ln\left(\frac{\mathcal{E} - vBl}{\mathcal{E}}\right)$$

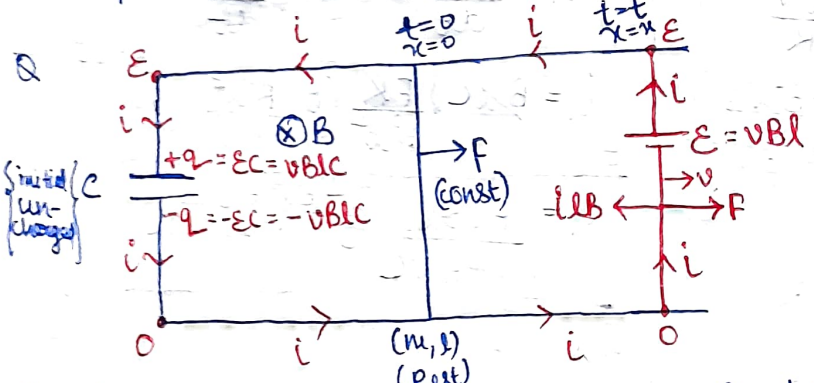
$$-\frac{B^2l^2t}{mR} = \ln\left(\frac{\mathcal{E} - vBl}{\mathcal{E}}\right)$$

$$\boxed{v = \frac{\mathcal{E}}{Bl}\left\{1 - e^{-\frac{B^2l^2t}{mR}}\right\}}$$

Now, $i = \frac{\mathcal{E} - vBl}{R} = \frac{\mathcal{E}}{R} - \frac{\mathcal{E}}{R}\left\{1 - e^{-\frac{B^2l^2t}{mR}}\right\} \Rightarrow \boxed{i = \frac{\mathcal{E}}{R}e^{-\frac{B^2l^2t}{mR}}}$



Q



At any time t,

① Find velocity of rod

② current in circuit / charge on capacitor

③ Displacement of rod

$$i = \frac{dq}{dt} = \frac{d}{dt}(vBlC) = BlC\frac{dv}{dt} \quad -(1)$$

$$F - ilB = m\frac{dv}{dt} \quad -(2)$$

$$\Rightarrow F - B^2l^2C\frac{dv}{dt} = m\frac{dv}{dt}$$

$$\Rightarrow F = \frac{dv}{dt}(m + B^2l^2C)$$

$$a = \frac{dv}{dt} = \frac{F}{m + B^2l^2C} = \text{const}$$

① $v = u + at \Rightarrow v = 0 + \frac{Ft}{m + B^2l^2C}$

$$\boxed{v = \frac{Ft}{m + B^2l^2C}}$$

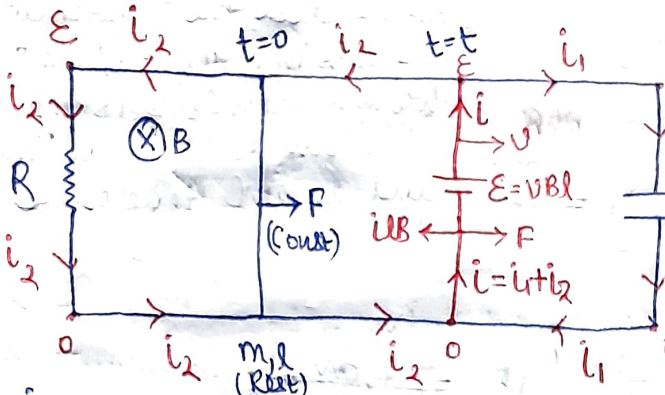
② $i = BlC\frac{dv}{dt} = \frac{BlCF}{m + B^2l^2C}$

$$q = \int i\,dt = it = \frac{BlCFt}{m + B^2l^2C}$$

③ $s = ut + \frac{1}{2}at^2$

$$x = 0 + \frac{1}{2}\left(\frac{Ft^2}{m + B^2l^2C}\right)$$

Q



Find,

① Velocity of rod

② Current through R

③ Charge on capacitor

④ Current through capacitor

$$i_1 = \frac{dq}{dt} = \frac{d}{dt}(vBlC) = BlC\frac{dv}{dt} \quad -①$$

$$i_2 = \frac{\varepsilon}{R} = \frac{vBl}{R} \quad -②$$

$$F - (ilB) = m\frac{dv}{dt} \quad -③$$

$$= F - (i_1 + i_2)lB = m\frac{dv}{dt}$$

$$= F - B^2l^2C\frac{dv}{dt} - \frac{B^2l^2v}{R} = m\frac{dv}{dt}$$

$$= F - \frac{B^2l^2v}{R} = (m + B^2l^2C)\frac{dv}{dt}$$

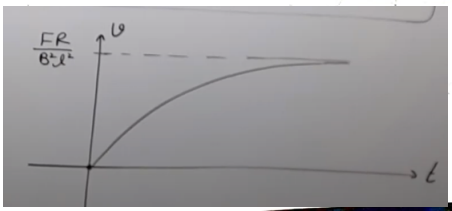
$$= \int_0^t dt = (m + B^2l^2C)\int_0^v \frac{dv}{F - \frac{B^2l^2v}{R}}$$

$$= t = \frac{(m + B^2l^2C)R}{-B^2l^2}\ln\left(\frac{F - \frac{B^2l^2v}{R}}{F}\right)$$

$$= \ln\left(\frac{F - \frac{B^2l^2v}{R}}{F}\right) = \frac{-B^2l^2t}{(m + B^2l^2C)R}$$

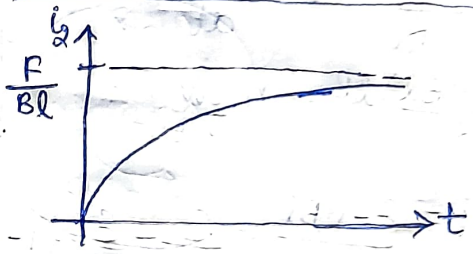
$$= 1 - \frac{B^2l^2v}{FR} = e^{\frac{-B^2l^2t}{(m + B^2l^2C)R}}$$

$$\boxed{v = \frac{FR}{B^2l^2}\left\{1 - e^{\frac{-B^2l^2t}{(m + B^2l^2C)R}}\right\}}$$



② $i_2 = \frac{\varepsilon}{R} = \frac{vBl}{R}$

$$\boxed{i_2 = \frac{F}{Bl}\left(1 - e^{\frac{-B^2l^2t}{(m + B^2l^2C)R}}\right)}$$



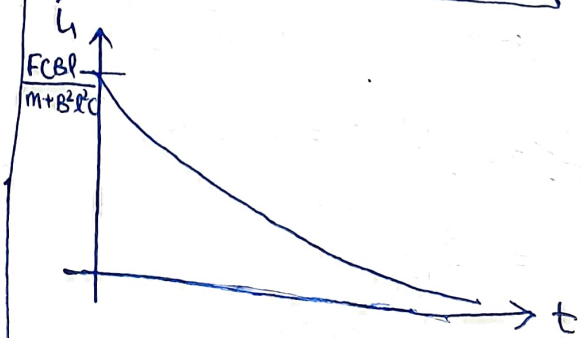
③ $q = \varepsilon C = vBlC$

$$\boxed{q = \frac{FRC}{Bl}\left\{1 - e^{\frac{-B^2l^2t}{(m + B^2l^2C)R}}\right\}}$$

④ $i_1 = \frac{dq}{dt} = BlC\cdot\frac{dv}{dt}$

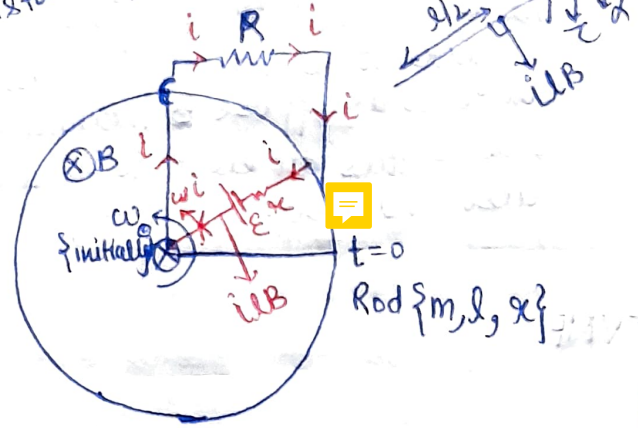
$$i_1 = BlC\left\{\frac{FR}{B^2l^2}\left(0 + \frac{B^2l^2}{(m + B^2l^2C)R}e^{\frac{-B^2l^2t}{(m + B^2l^2C)R}}\right)\right\}$$

$$\boxed{i_1 = \frac{FCBl}{(m + B^2l^2C)}e^{\frac{-B^2l^2t}{(m + B^2l^2C)R}}}$$



Q Horizontal

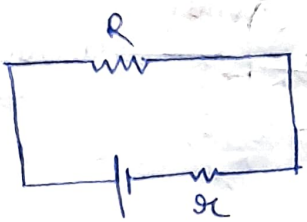
Resistance of loop = 0



At any time t,

① Find ω of rod ② Current

③ Power of R ④ H_R



$\mathcal{E} = \frac{B\omega l^2}{2}$

$i = \frac{\mathcal{E}}{R_{eq}} = \frac{B\omega l^2}{2(R+r)}$ — ①

① Now, $\tau = I\alpha$

$\tau = i l B\left(\frac{l}{2}\right) = \frac{ml^2}{3}\alpha$

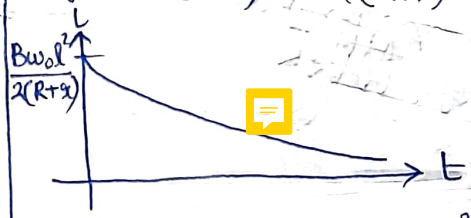
$\alpha = \frac{3iB}{2m} = \frac{3B^2l^2\omega}{4m(R+r)} = \frac{-d\omega}{dt}$

$= \frac{-3B^2l^2}{4m(R+r)}\int_0^t dt = \int_{\omega_0}^{\omega}\frac{d\omega}{\omega}$

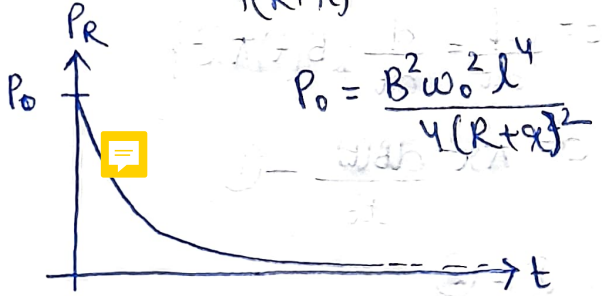
$= \frac{-3B^2l^2t}{4m(R+r)} = \ln\left(\frac{\omega}{\omega_0}\right)$

$= \boxed{\omega = \omega_0 e^{\frac{-3B^2l^2t}{4m(R+r)}}}$

② $i = \frac{B\omega l^2}{2(R+r)} = \frac{B\omega_0 l^2}{2(R+r)} e^{\frac{-3B^2l^2t}{4m(R+r)}}$



③ $P_R = i^2R = \frac{B^2\omega_0^2 l^4}{4(R+r)^2} e^{\frac{-3B^2l^2t}{2m(R+r)}}$



$P_0 = \frac{B^2\omega_0^2 l^4}{4(R+r)^2}$

Area under P_R v/s t gives H_R.

④ Method 1: $H_R = \int_0^\infty P_R\,dt$

Method 2: Rotational KE of rod gets dissipated as Heat.

$\frac{H_R}{H_r} = \frac{R}{r}$ — ①

$\frac{1}{2}\left(\frac{ml^2}{3}\right)\omega_0^2 = H_R + H_r$

$= \frac{1}{2}\left(\frac{ml^2}{3}\right)\omega_0^2 = H_R\left(\frac{R+r}{R}\right)$

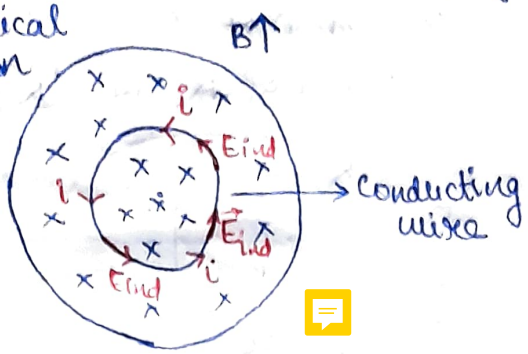
$\Rightarrow H_R = \frac{1}{2}\left(\frac{ml^2\omega_0^2}{3}\right)\frac{R}{R+R}$

$\boxed{H_R = \frac{ml^2\omega_0^2 R}{6(R+r)}}$

Electric Field due to time varying magnetic field (TVMF):

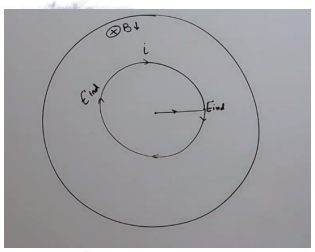
Cylindrical Region

ex1.



As B↑ in cylindrical region, a current is induced in conducting wire so as oppose B increasing B. Due to this an electric field is also induced.

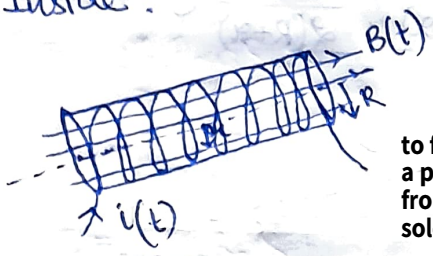
ex2.



→ Electric field is induced by TVMF
→ They make circular loops
→ Centre: Axis of cylinder

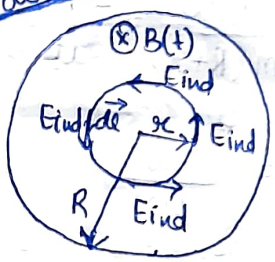
Formula of induced Electric Field:

(i) Inside:



to find induced EF at a point P at a distance r from the axis of the solenoid

Side View



By Faradays law:

induced EMF

$$\mathcal{E} = \frac{d\phi}{dt} = \frac{d}{dt}\{B(t)\pi r^2\}$$

$$\mathcal{E} = \pi r^2 \frac{dB(t)}{dt} \quad -(1)$$

By definition:

$$\mathcal{E} = \frac{W}{q} = \frac{\oint q\vec{E}_{ind}\cdot\vec{dl}}{q} = \oint \vec{E}_{ind}\cdot\vec{dl}$$

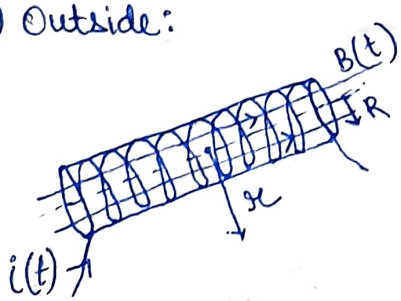
$$\mathcal{E} = \oint E_{ind}\, dl \cos 0^\circ = E_{ind}\oint dl$$

$$\mathcal{E} = E_{ind}(2\pi r) \quad -(2)$$

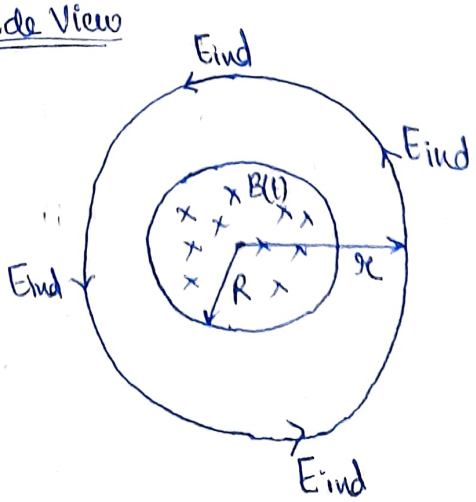
Now, (1) = (2)

$$\pi r^2 \frac{dB}{dt} = E_{ind}(2\pi r) \Rightarrow \boxed{E_{ind} = \frac{r}{2}\frac{dB}{dt}} \quad \boxed{E \propto r}$$

(ii) Outside:



Side View



By Faraday's law:

$$\mathcal{E} = \frac{d\phi}{dt} = \frac{d}{dt}\{B(t) \pi R^2\}$$

$$\mathcal{E} = \pi R^2 \frac{dB}{dt} \quad -(1)$$

By definition:

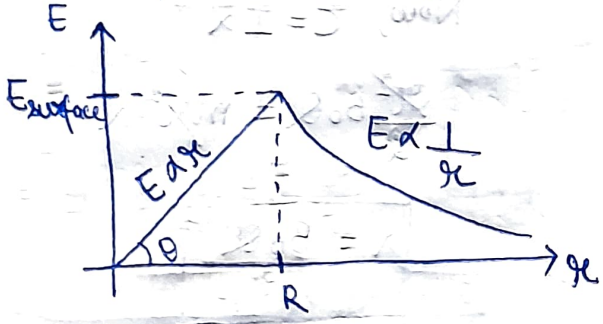
$$\mathcal{E} = \frac{W}{q} = \frac{\oint q \cdot \vec{E}_{ind} \cdot d\vec{l}}{q} = \oint \vec{E}_{ind} \cdot d\vec{l}$$

$$\mathcal{E} = E_{ind} \oint dl = E_{ind}(2\pi r) \quad -(2)$$

Now, (1) = (2)

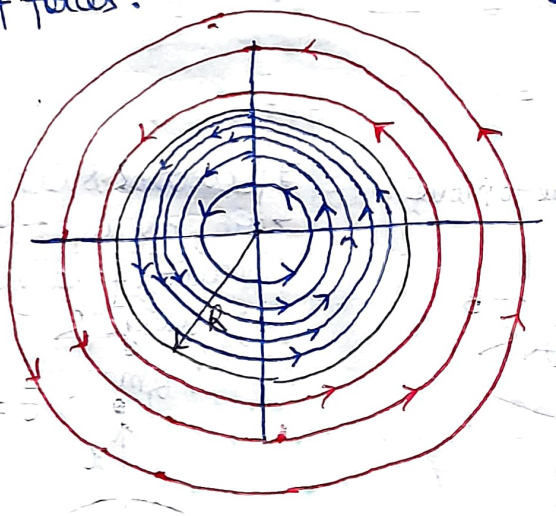
$$\pi R^2 \frac{dB}{dt} = E_{ind}(2\pi r) \Rightarrow \boxed{E_{ind} = \frac{R^2}{2r}\frac{dB}{dt}} \qquad \boxed{E_{ind} \propto \frac{1}{r}}$$

→ Maximum electric field is at surface, $E_{surface} = \frac{R}{2}\frac{dB}{dt}$



$$\boxed{\tan\theta = \frac{1}{2}\left(\frac{dB}{dt}\right)}$$

Electric lines of forces:



Inside cylinder: ⊗ B ↑

$r < R$: $E \propto r$

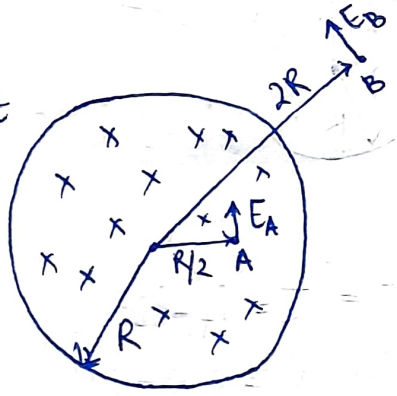
⇒ More line density

$r > R$: $E \propto \frac{1}{r}$

→ less line density

Q

$B = ct$



Find electric field at point A and B

$$E_A = \frac{r}{2}\frac{dB}{dt} = \frac{R/2}{2}\frac{d}{dt}(ct) = \frac{Rc}{4}$$

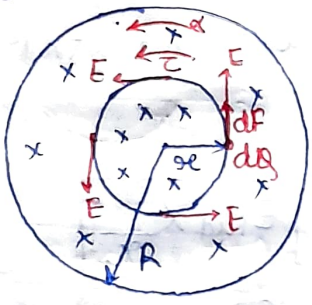
$$E_B = \frac{R^2}{2r}\frac{dB}{dt} = \frac{R^2}{2(2R)}\frac{d}{dt}(ct)$$

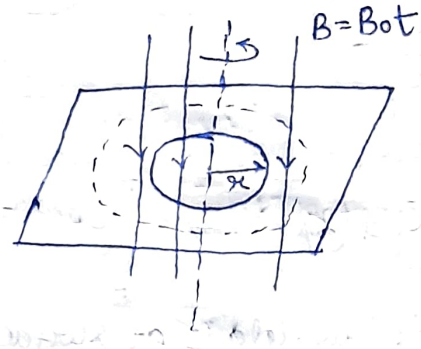
$$E_B = \frac{Rc}{4}$$

Q Find torque on the charged ring. Find its angular acceleration.

Given ring is uniform (m, r), non-conducting with charge Q uniformly distributed

$B = B_0 t$





$$E = \frac{r}{2}\frac{d}{dt}(B_0 t) = \frac{r}{2}B_0$$

$$dF = E\,dQ = \frac{r}{2}B_0\,dQ$$

$$d\tau = \frac{r}{2}B_0\,dQ\,r$$

$$\tau = \int d\tau = \frac{r^2}{2}B_0\int dQ$$

$$\boxed{\tau = \frac{r^2}{2}B_0 Q}$$

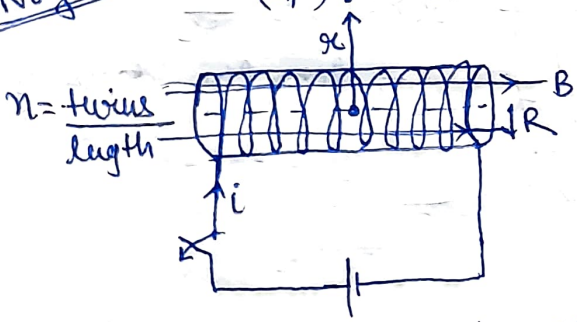
Now, $\tau = I\alpha$

$$\frac{r^2}{2}B_0 Q = mr^2\alpha$$

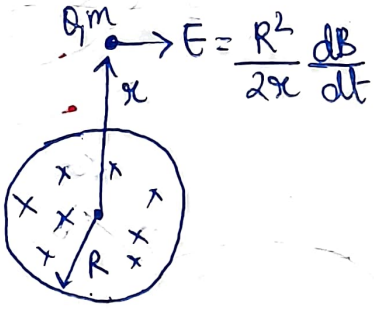
$$\boxed{\alpha = \frac{B_0 Q}{2m}}$$

Q No gravity

(Q, m). Rest, free to move



If current suddenly becomes zero, as we switch off the circuit, find velocity achieved by charge Q.



As we switch off the circuit, current becomes zero in dt time. In the same dt time B also becomes zero from $\mu_0 ni$. Due to this change in B, particle will experience an electric field **only during that dt time**

$$F = EQ = \frac{R^2 Q}{2r}\frac{dB}{dt}$$

$$\Rightarrow \int F\,dt = \frac{R^2 Q}{2r}\int_0^{\mu_0 ni} dB$$

$$J = \frac{R^2 Q}{2r}(\mu_0 ni)$$

Using IMT $\Rightarrow J = P_f - P_i$

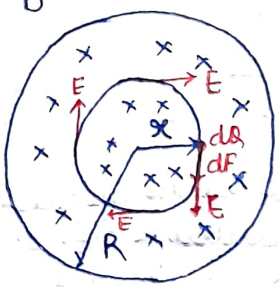
$$\frac{R^2 Q}{2r}(\mu_0 ni) = mv - 0$$

$$\boxed{v = \frac{R^2 Q}{2mr}(\mu_0 ni)}$$

Q

Given uniform ring (m, r), non-conducting (Q). B dies and becomes zero instantaneously. Find angular velocity achieved by ring.

B = uniform



$dF = E\,dQ = \frac{r\,dB}{2\,dt}\,dQ$

$d\tau = E\,dQ\,r$

$\tau = \int d\tau = Er\int dQ = ErQ$

$\tau = \frac{r^2 Q}{2}\frac{dB}{dt}$

$\therefore \boxed{\omega = \frac{QB}{2m}}$

Now, ω

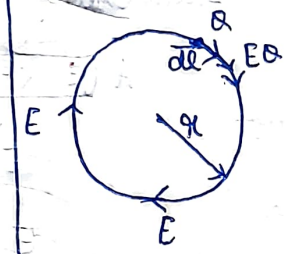
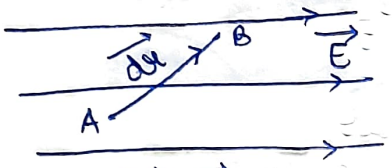
$\int \tau\,dt = \frac{r^2 Q}{2}\int_B^0 dB$

$\int \tau\,dt = \frac{r^2}{2} Q|B| = \frac{r^2}{2} QB$

using AIAMT $\Rightarrow A \cdot I = L_f - L_i$

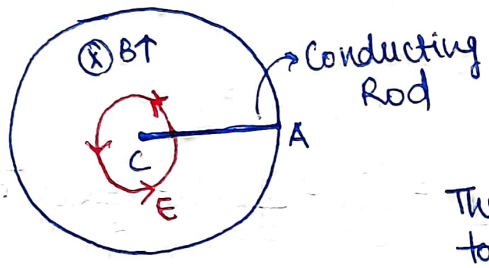
$\frac{r^2}{2} QB = m r^2 \omega$

Comparison:

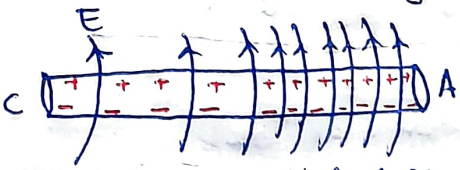
$\vec{E}$ due to Charges	$\vec{E}$ due to TVMF
→ Conservative Field	→ ELOF can make closed loop
→ Potential defined	 $dW = EQ\,dl$, $W = EQ\int dl$, $W = EQ2\pi r \neq 0$
→ Cannot form closed loop	→ Non-conservative field
 $V_B - V_A = -\vec{E}\cdot\vec{dr}$, $\boxed{dV = -\vec{E}\cdot\vec{dr}}$	→ No potential or potential energy is defined due to this EF

**

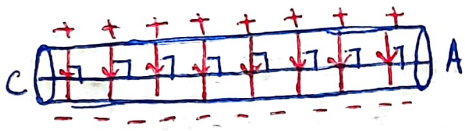
Q



which point is at higher potential?



There will be no potential defined for E due to TVMF. But due to this electric field, charges will be induced in upper and lower surface of rod. An $\vec{E}_{ind}$ develops due to charges for which potential is defined.



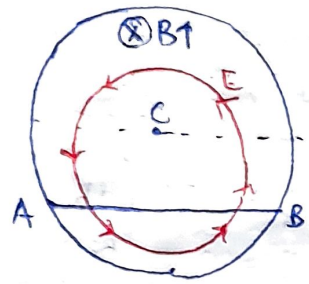
Line joining AC always cuts the electric field at 90°.

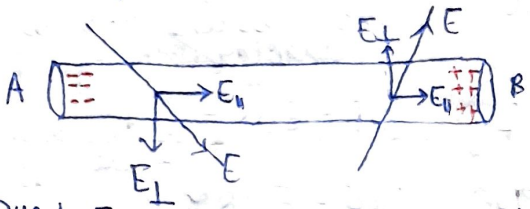
∴ It is an equipotential surface

$\therefore \boxed{V_A = V_C}$

Q**

Which point is at higher potential A or B?



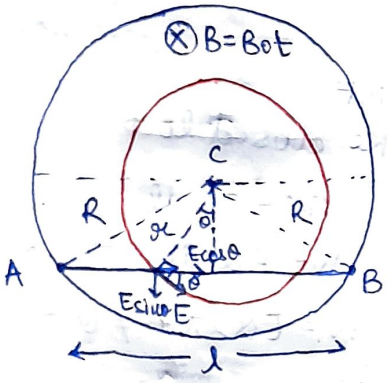


Due to $E_{\parallel}$, charges gets induced and an induced E.F develops. $E_{\perp}$ also develope E.F. but work done due to it is zero for A B.

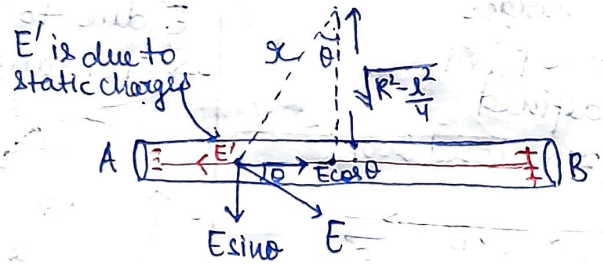
Electric field is from higher potential to lower potential.

$\therefore \boxed{V_B > V_A}$

Q*



Find $V_B - V_A$

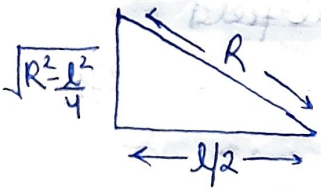


in electrostatic condition

$E' = E\cos\theta$

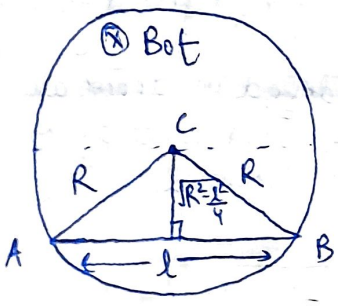
$\Rightarrow E' = \frac{x}{2} B_0 \left(\frac{\sqrt{R^2 - \frac{l^2}{4}}}{x} \right)$

$\Rightarrow \boxed{E' = \frac{B_0}{2}\sqrt{R^2 - \frac{l^2}{4}}}$ {const}



Now, $\boxed{V_B - V_A = E'l = \frac{B_0 l}{2}\sqrt{R^2 - \frac{l^2}{4}}}$

Alternate



In loop ABC, $\mathcal{E}_{ABC} = \frac{d\phi}{dt} = \frac{d}{dt}\left\{ B_0 t \times \frac{1}{2} \times l \times \sqrt{R^2 - \frac{l^2}{4}} \right\}$

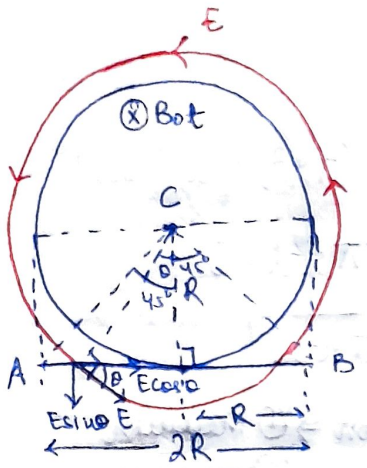
$\mathcal{E}_{ABC} = \frac{B_0 l}{2}\sqrt{R^2 - \frac{l^2}{4}}$

Now, $\mathcal{E}_{ABC} = \mathcal{E}_{AB} + \mathcal{E}_{BC} + \mathcal{E}_{CA}$

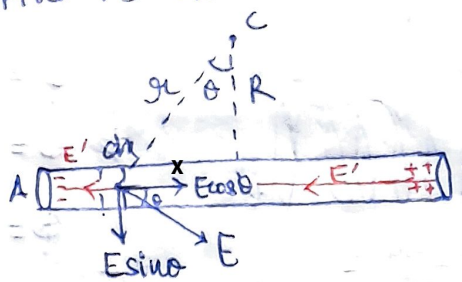
As $V_C = V_A$ and $V_C = V_B \Rightarrow \mathcal{E}_{BC} = \mathcal{E}_{CA} = 0$

$\boxed{\mathcal{E}_{AB} = \frac{B_0 l}{2}\sqrt{R^2 - \frac{l^2}{4}}}$

Q



Find $V_B - V_A$



$\frac{x}{R} = \tan\theta$

$x = R\tan\theta$

$dx = R\sec^2\theta\, d\theta$

$dx = R\frac{x^2}{R^2}\, d\theta$

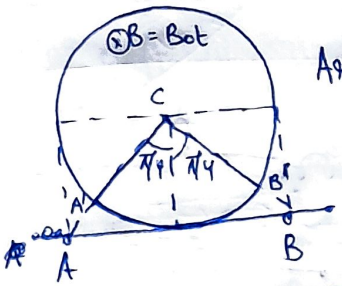
$E' = E\cos\theta = \frac{R^2}{2x} B_0 \left(\frac{R}{x}\right)$

$$\boxed{E' = \frac{R^3 B_0}{2x^2}} \quad \{\text{Not const}\}$$

$$dV = E'dx = \frac{R^3 B_0\, dx}{2x^2} = \frac{R^3 B_0}{2x^2} R\frac{x^2}{R^2} d\theta = \frac{B_0 R^2}{2} d\theta$$

$$\Delta V = \int dV = \frac{B_0 R^2}{2} \int_{-\pi/4}^{\pi/4} d\theta = \frac{B_0 R^2}{2} (\theta)_{-\pi/4}^{\pi/4} = \frac{B_0 R^2}{2}\left\{\frac{\pi}{4} - \left(-\frac{\pi}{4}\right)\right\} = \frac{B_0 R^2 \pi}{4}$$

Alternate:



Area of $A'B'C = \frac{R^2\theta}{2}$

$= \frac{R^2 \pi/2}{2}$

$A = \frac{R^2\pi}{4}$

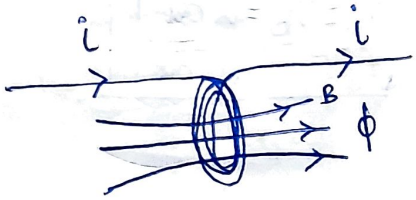
$\mathcal{E}_{ABC} = \frac{d}{dt}\{\phi_{ABC}\}$

$= \frac{d}{dt}\{B_0 t \frac{\pi R^2}{4}\}$

$= \frac{B_0 \pi R^2}{4} = \mathcal{E}_{AB} + \mathcal{E}_{BC}^{0} + \mathcal{E}_{CA}^{0}$

$$\therefore \boxed{\mathcal{E}_{AB} = \frac{B_0 \pi R^2}{4}}$$

Self-inductance:



$$\boxed{\phi \propto i}$$

$\Rightarrow \phi = Li \Rightarrow \boxed{L = \frac{\phi}{i}}$

→ Flux through a current carrying loop is directly proportional to current through the loop and constant of proportionality is coefficient of self-inductance.

L = Coefficient of self-inductance

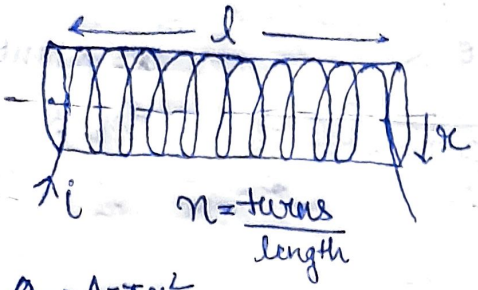
→ Scalar

→ Non-negative

→ unit = $\frac{\text{Weber}}{\text{Amp}}$ = Henry (H)

→ Dimension = $[ML^2 I^{-2} T^{-2}]$

Self-inductance of an ideal solenoid:

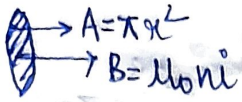


$\phi = Li$

$\Rightarrow \phi = (\mu_0 ni)(\pi r^2)(nl)(\cos 0^\circ)$

$\phi = \{\mu_0 n^2 \pi r^2 l\} i$

$\therefore \boxed{L = \mu_0 n^2 \pi r^2 l}$



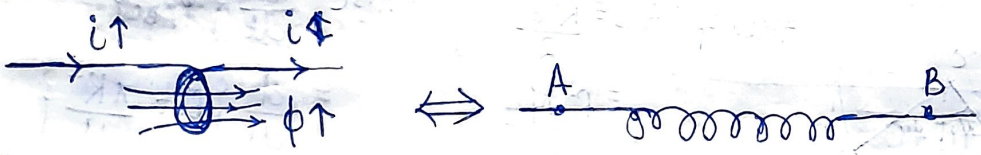
L depends on ⇒ ① Medium
② No of turns
③ Shape, size and geometry

$$\frac{\text{Self-inductance}}{\text{Volume}} = \mu_0 n^2$$

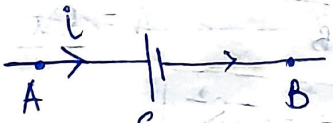
Inductor:

Again, self-inductance: alternate definition

Case ①:



By lenz law, emf induces to oppose change in flux



At time = t, $i \to \phi = Li$

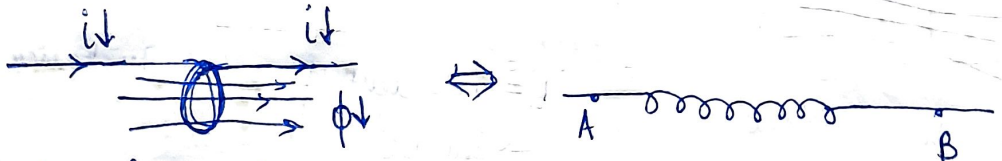
$t = t + dt$, $i + di \to \phi + d\phi = Li + Ldi$

$\mathcal{E}_{ind} = \left|\frac{d\phi}{dt}\right|$

$\mathcal{E}_{net} = \left|\frac{d\phi}{dt}\right| = \left|L\frac{di}{dt}\right|$

$\therefore \boxed{V_A - V_B = \left|L\frac{di}{dt}\right|}$

Case ②:



By lenz law, emf induces to oppose change in flux.

$\mathcal{E}_{ind} = \left|\frac{d\phi}{dt}\right| = \left|L\frac{di}{dt}\right|$

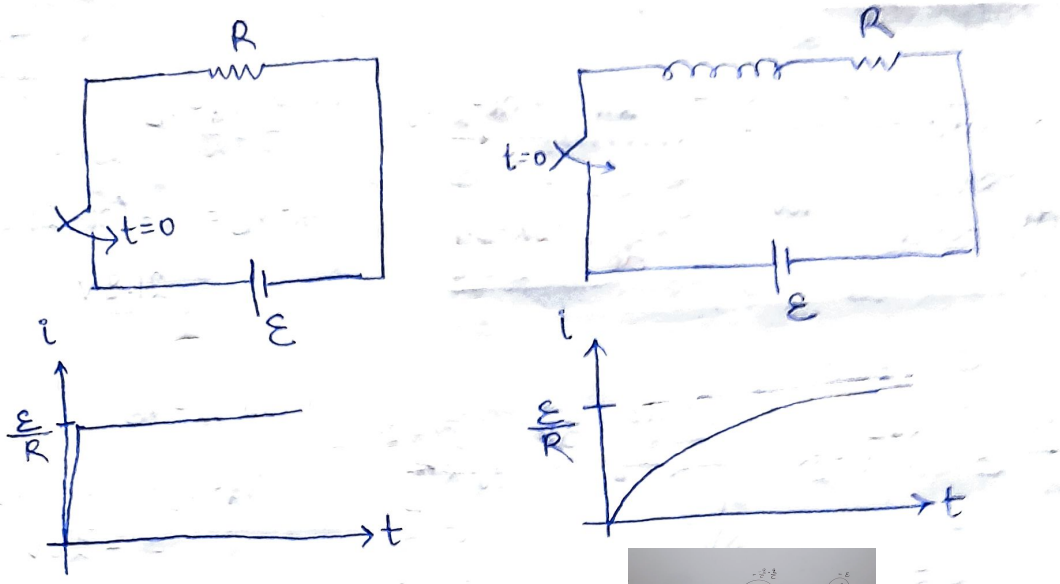
$\boxed{V_B > V_A}$

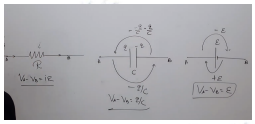
$\therefore \boxed{V_B - V_A = \left|L\frac{di}{dt}\right|}$

→ Self inductance is property of loop which makes sure that magnetic flux and current in loop doesn't change instantaneously.

→ Self-inductance is property of loop due to which loop tries to maintain constant flux and current.

Note: Above is valid only if L is fixed, which is in general.





Potential difference across inductor:

①

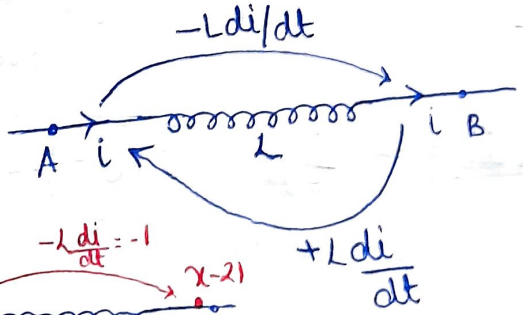
$$V_A - V_B = L\frac{di}{dt}$$

* Use $\frac{di}{dt}$ with sign.

+ve if increasing, -ve if decreasing.

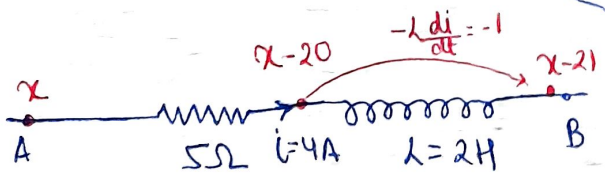
Replace inductor with cell of emf $\left|L\frac{di}{dt}\right|$ and decide polarity using Lenz law.

②



Use $\frac{di}{dt}$ with sign

Q



$\frac{di}{dt} = 0.5 \frac{Amp}{sec}$

Find $V_A - V_B$

$V_A - V_B = x - (x-21) = 21V$

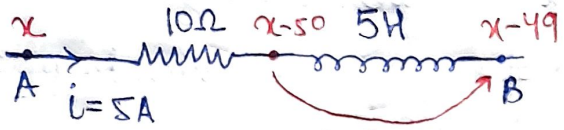
Method 2:

$V_A - V_B = 21V$

$\mathcal{E} = \left|L\frac{di}{dt}\right| = 1V$

Q

Find $V_A - V_B$



$\frac{di}{dt} = -0.2 \frac{Amp}{sec}$

$-L\frac{di}{dt} = -5x - 0.2$

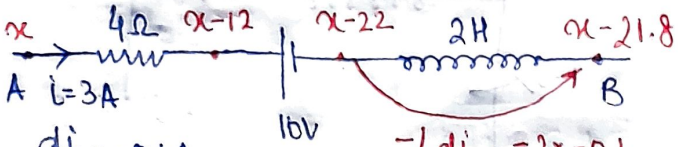
$= 1$

$V_A - V_B = x - (x-49)$

$= 49V$

Q

Find $V_A - V_B$



$\frac{di}{dt} = -0.1 \frac{A}{s}$

$-L\frac{di}{dt} = -2x - 0.1$

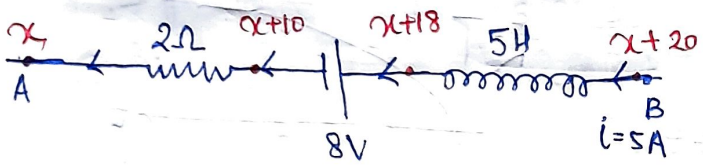
$= 0.2$

$V_A - V_B = x - (x - 21.8)$

$= 21.8V$

Q

Find $V_A - V_B$



$\frac{di}{dt} = 0.4 \frac{Amp}{sec}$

$V_A - V_B = x - (x+20)$

$= -20V$

Energy stored in an inductor:

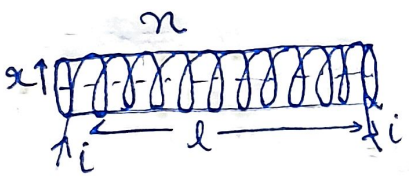
$P = Vi = \left(L\frac{di}{dt}\right) i$

$U = \int P dt = L\int_0^i i\,di = L\left(\frac{i^2}{2}\right)_0^i$

$\therefore \boxed{U = \frac{1}{2}Li^2}$

Load → Stores electrical energy and converts in magnetic energy.

Energy stored in an ideal solenoid:



$U = \frac{1}{2}Li^2 = \frac{1}{2}\left(\mu_0 n^2 \pi x^2 l\right) i^2$

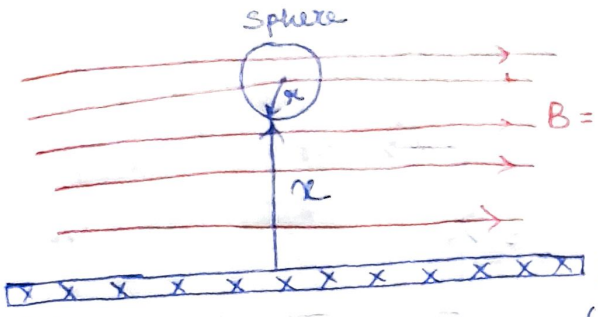
Energy density $= \frac{U}{Vol.} = \frac{1}{2}\mu_0 n^2 i^2$

$= \frac{(\mu_0 n i)^2}{2\mu_0}$

$= \frac{B^2}{2\mu_0}$

$\therefore \boxed{\text{Energy density} = \frac{dU}{dV} = \frac{B^2}{2\mu_0}}$

Q



$K = \frac{\text{amp}}{\text{meter}}$

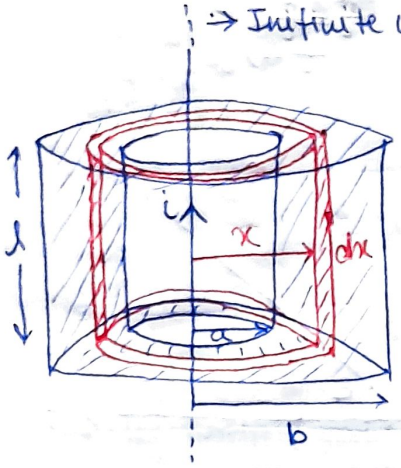
Find magnetic energy stored in sphere

$$\frac{dU}{dV} = \frac{B^2}{2\mu_0}$$

$$\Rightarrow U = \frac{B^2}{2\mu_0}\left(\frac{4}{3}\pi x^3\right)$$

$$\boxed{U = \frac{\left(\frac{\mu_0 K}{2}\right)^2}{2\mu_0}\left(\frac{4}{3}\pi x^3\right)}$$

Q



Find magnetic energy stored in the shaded region.

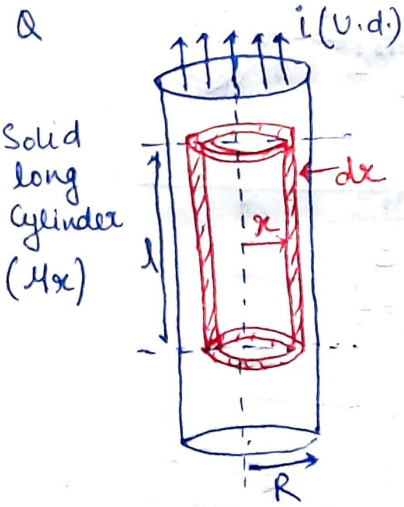
$$B = \frac{\mu_0 i}{2\pi x}$$

$$\frac{dU}{dV} = \frac{B^2}{2\mu_0} = \frac{\mu_0^2 i^2}{4\pi^2 x^2 \, 2\mu_0} = \frac{\mu_0 i^2}{8\pi^2 x^2}$$

$$dU = \frac{\mu_0 i^2}{8\pi^2 x^2} \cdot 2\pi x\, dx\, l = \frac{\mu_0 i^2 l}{4\pi}\frac{dx}{x}$$

$$U = \int dU = \frac{\mu_0 i^2 l}{4\pi}\int_a^b \frac{dx}{x} = \frac{\mu_0 i^2 l}{4\pi}\ln\left(\frac{b}{a}\right)$$

Q



Find magnetic energy stored per unit length inside this solid cylinder.

$$B = \frac{\mu_0 \mu_r J x}{2} = \frac{\mu_0 \mu_r}{2}\frac{i}{\pi R^2}x = \frac{\mu_0 \mu_r i x}{2\pi R^2}$$

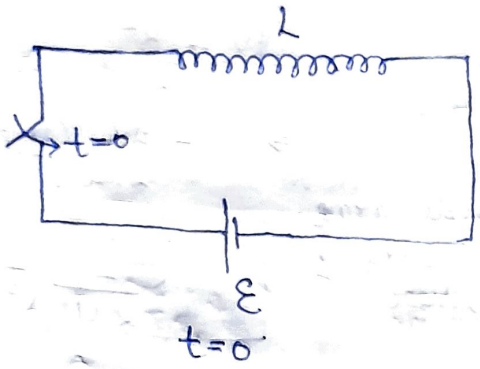
$$\frac{dU}{dV} = \frac{B^2}{2\mu_0\mu_r} = \frac{\mu_0^2 \mu_r^2 i^2 x^2}{4\pi^2 R^4 \, 2\mu_0\mu_r} = \frac{\mu_0 \mu_r i^2 x^2}{8\pi^2 R^4}$$

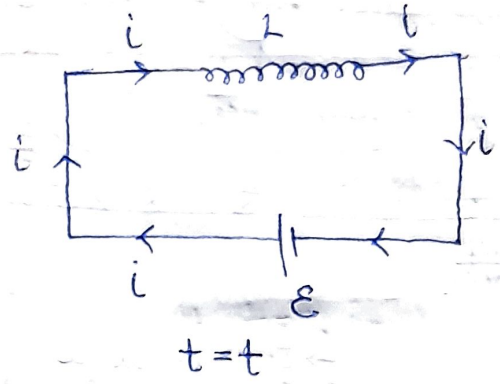
$$dU = \frac{\mu_0 \mu_r i^2 x^2}{8\pi^2 R^4} \cdot 2\pi x\, dx\, l = \frac{\mu_0 \mu_r i^2 l}{4\pi R^4} x^3 dx$$

$$U = \int dU = \frac{\mu_0 \mu_r i^2 l}{4\pi R^4}\int_0^R x^3 dx = \frac{\mu_0 \mu_r i^2 l}{4\pi R^4}\cdot\left(\frac{R^4}{4}\right)$$

$$U = \frac{\mu_0 \mu_r i^2 l}{16\pi} \Rightarrow \boxed{\frac{U}{l} = \frac{\mu_0 \mu_r i^2}{16\pi}}$$

Circuit Solutions :





Using KVL:

$$\varepsilon - L\frac{di}{dt} = 0$$

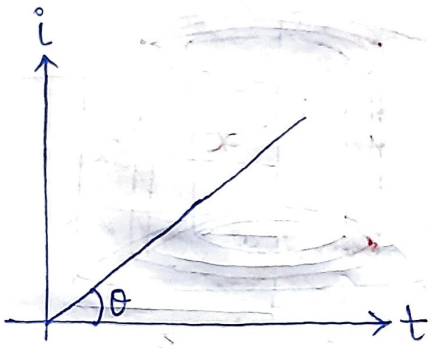
$$\Rightarrow \varepsilon = L\frac{di}{dt}$$

$$\Rightarrow \int_0^t \varepsilon\, dt = L\int_0^i di$$

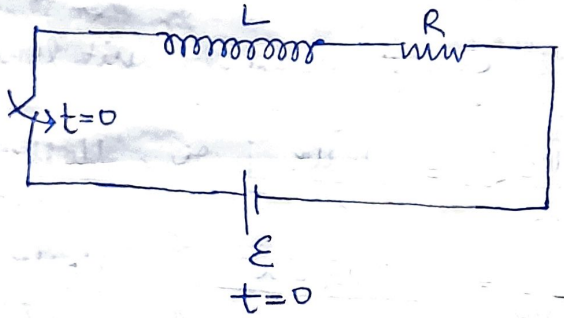
$$\Rightarrow \varepsilon t = Li$$

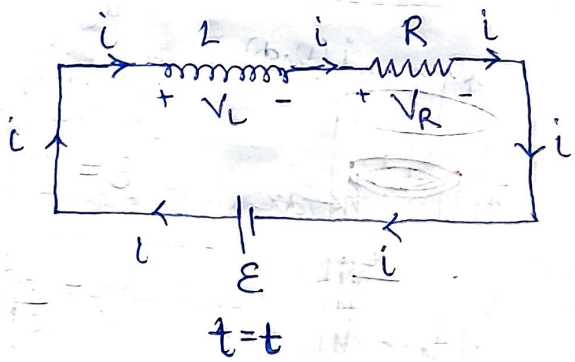
$$\therefore \boxed{i = \frac{\varepsilon}{L}t}$$

$$\boxed{\tan\theta = \frac{\varepsilon}{L}}$$



Growth of current in LR circuit:





(I) Current at any time t.

using KVL:

$$\varepsilon - L\frac{di}{dt} - iR = 0$$

$$\Rightarrow \varepsilon - iR = L\frac{di}{dt}$$

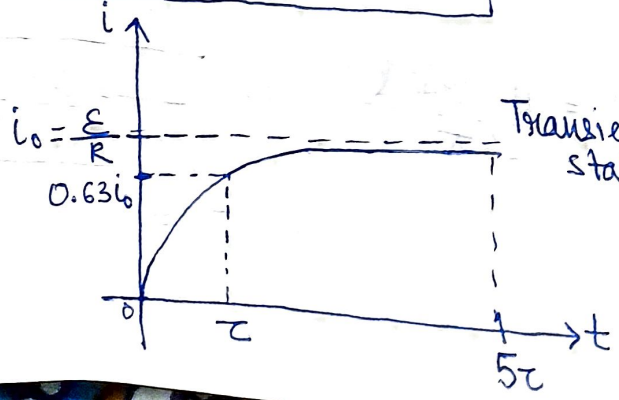
$$\Rightarrow \int_0^t dt = L\int_0^i \frac{di}{\varepsilon - iR}$$

$$\Rightarrow t = \frac{L}{-R}\ln\left(\frac{\varepsilon - iR}{\varepsilon}\right)$$

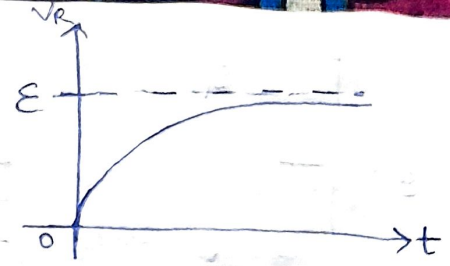
$$\Rightarrow \ln\left(\frac{\varepsilon - iR}{\varepsilon}\right) = \frac{-Rt}{L}$$

$$\Rightarrow \boxed{i = \frac{\varepsilon}{R}\left(1 - e^{-\frac{Rt}{L}}\right)} \qquad \boxed{i = \frac{\varepsilon}{R}\left(1 - e^{-t/\tau}\right)}$$

$$\boxed{\tau = \frac{L}{R} = \text{Time constant}}$$

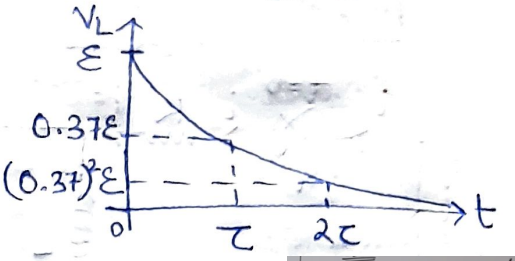


② $V_R = iR = \mathcal{E}(1-e^{-Rt/L}) = \mathcal{E}(1-e^{-t/\tau})$



③ $V_L = L\frac{di}{dt} = \mathcal{E} - V_R = \mathcal{E}e^{-Rt/L}$

$$\boxed{V_L = \mathcal{E}e^{-t/\tau}}$$



④ $P_R = i^2R = \left[\frac{\mathcal{E}}{R}(1-e^{-Rt/L})\right]^2 R$

$$\boxed{P_R = \frac{\mathcal{E}^2}{R}\{1-e^{-Rt/L}\}^2}$$

⑤ $P_L = V_L i = \mathcal{E}e^{-t/\tau}\cdot\frac{\mathcal{E}}{R}(1-e^{-t/\tau})$

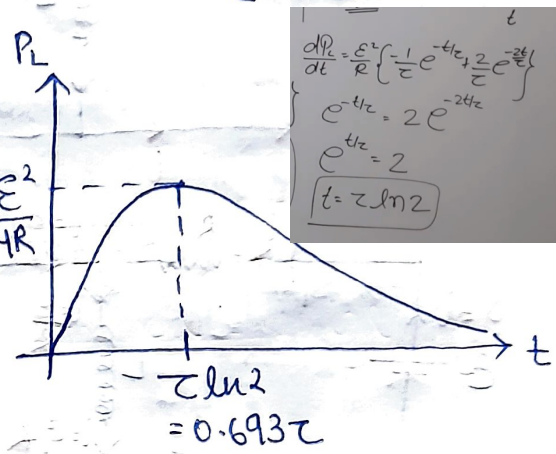
$$\boxed{P_L = \frac{\mathcal{E}^2}{R}\{e^{-t/\tau} - e^{-2t/\tau}\}}$$

$\frac{dP_L}{dt} = \frac{\mathcal{E}^2}{R}\left\{-\frac{1}{\tau}e^{-t/\tau} + \frac{2}{\tau}e^{-2t/\tau}\right\}$

$e^{-t/\tau} = 2e^{-2t/\tau}$

$e^{t/\tau} = 2$

$t = \tau \ln 2$



Area under P_L v/s t curve gives energy stored by inductor.

Area $= U = \frac{1}{2}Li_{\infty}^2 = \frac{1}{2}L\left(\frac{\mathcal{E}}{R}\right)^2 = \frac{L\mathcal{E}^2}{2R^2}$

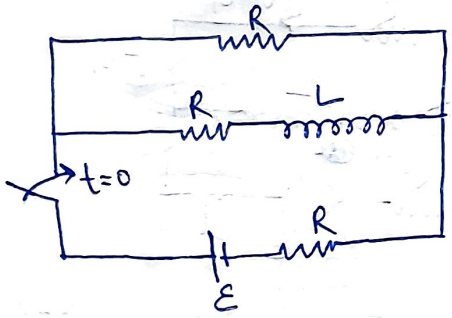
Results:

① At $t=0$, if we switch on the circuit, then current will still be zero and inductor will behave like broken wire.

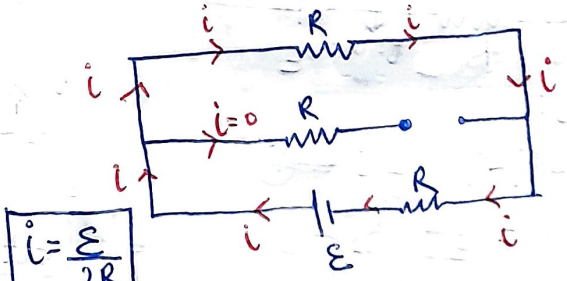
② At steady state, ie $t\to\infty$, inductor will behave like short wire.

③ In general, L = fix, so magnetic flux and current passing through an inductor will not change suddenly.

Q

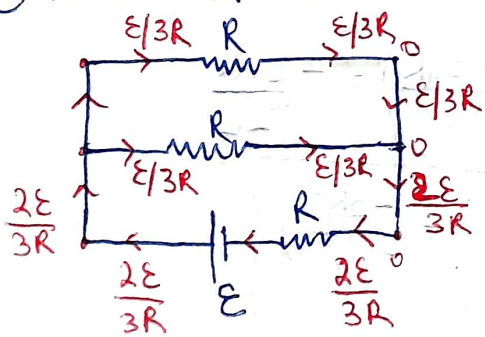


① At $t=0$, find current in each wire.



$$\boxed{i = \frac{\mathcal{E}}{2R}}$$

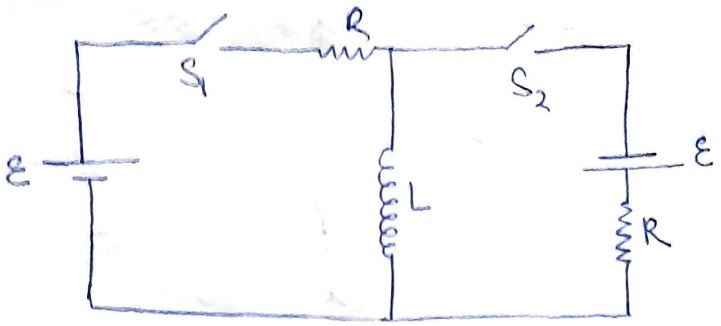
② At $t\to\infty$, find current in each wire.



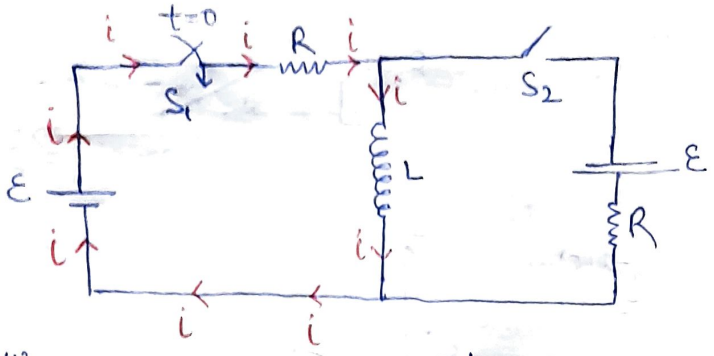
$$R_{eq} = \frac{R}{2} + R = \frac{3R}{2}$$

$$i = \frac{\mathcal{E}}{R_{eq}} = \frac{2\mathcal{E}}{3R}$$

Q



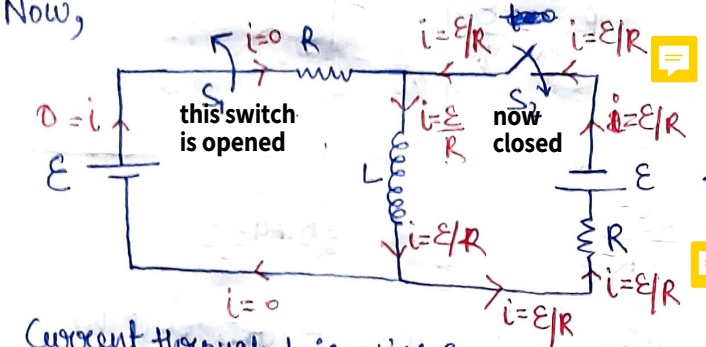
First S_1 is closed, find current in circuit. After a long time, S_1 is opened and S_2 is closed simultaneously, find current in circuit.



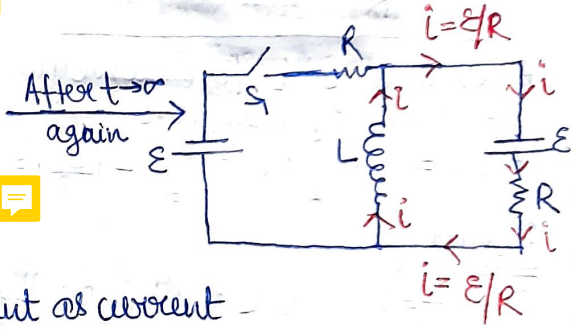
At, $t=0$, $i=0$

$t \to \infty$, $i = \frac{\varepsilon}{R}$

Now,

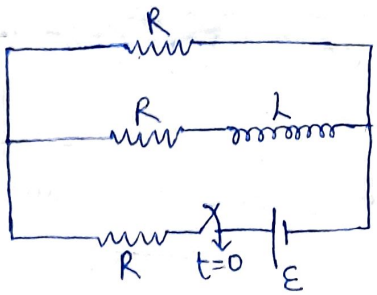


After $t \to \infty$ again

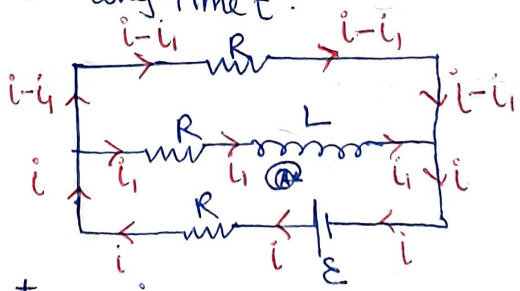


Current through L is still $\frac{\varepsilon}{R}$ at this instant as current in inductor does not change instantly.

Q*



Find current through inductor at any time t.



Using KVL:

$$\varepsilon - iR - (i - i_1)R = 0 \quad \text{--- (1)}$$

$$\Rightarrow iR = \varepsilon + i_1 R - \text{(1)}$$

$$= \frac{\varepsilon + i_1 R}{2}$$

In loop (A)

$$\varepsilon - iR - i_1 R - L\frac{di_1}{dt} = 0$$

From (1)

$$\Rightarrow \varepsilon - \frac{\varepsilon}{2} - \frac{i_1 R}{2} - i_1 R = L\frac{di_1}{dt}$$

$$\frac{\varepsilon - 3i_1 R}{2L} = \frac{di_1}{dt}$$

$$= \int_0^t \frac{dt}{2L} = \int_0^{i_1} \frac{di_1}{\varepsilon - 3i_1 R} \Rightarrow \frac{t}{2L} = \frac{-1}{3R} \ln\left(\frac{\varepsilon - 3i_1 R}{\varepsilon}\right)$$

$$= \ln\left(\frac{\varepsilon - 3i_1 R}{\varepsilon}\right) = -\frac{3Rt}{2L}$$

$$\boxed{i_1 = \frac{\varepsilon}{3R}\left(1 - e^{-\frac{3Rt}{2L}}\right)}$$

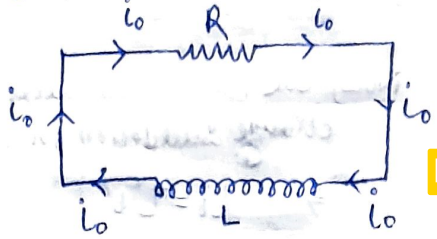
Trick: Only valid when initially current is zero in inductor.

$i = i_{ss}(1 - e^{-t/\tau})$

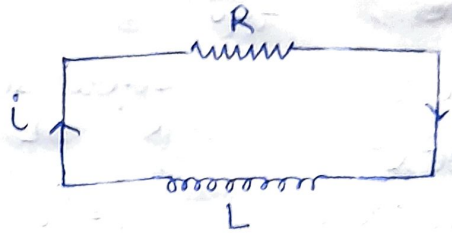
i_{ss} = Current at steady state

$\tau = \frac{L}{R_{eq}}$ where R_{eq} is across inductor when we replace battery by its internal resistance.

Decay of Current:



$t = 0$



$t = t$

① Current at any time t:

Using KVL:

$$-L\frac{di}{dt} - iR = 0$$

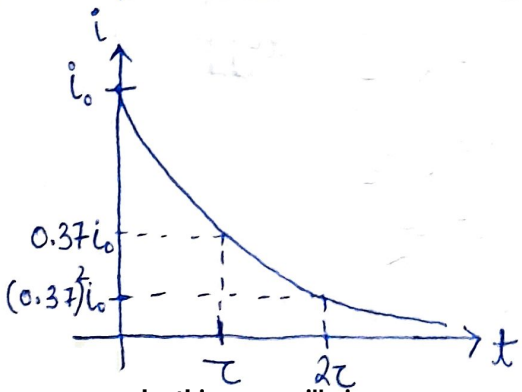
$$= -L\frac{di}{dt} = iR$$

$$= -\frac{L}{R}\int_{i_0}^{i}\frac{di}{i} = \int_0^t dt$$

$$= -\frac{L}{R}\ln\left(\frac{i}{i_0}\right) = t$$

$$= \boxed{i = i_0 e^{-Rt/L}}$$

$$\boxed{i = i_0 e^{-t/\tau}}$$



area under this curve will give the the total charge flown through the circuit

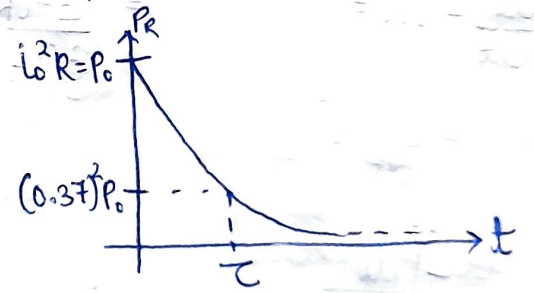
② Charge flow = $Q = \int_0^\infty i\,dt$

$$Q = i_0\int_0^\infty e^{-Rt/L}\,dt = \frac{i_0 L}{-R}\left(e^{-Rt/L}\right)_0^\infty$$

$$Q = \frac{i_0 L}{-R}\left(e^{-\infty} - e^{0}\right) = \frac{i_0 L}{-R}(0-1)$$

$$\boxed{Q = \frac{i_0 L}{R}}$$

③ $P_R = i^2 R = i_0^2 e^{-2Rt/L} R = i_0^2 R e^{-2t/\tau}$



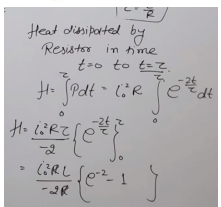
④ Heat dissipated by resistor:

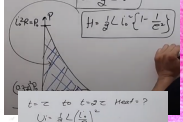
$$H_R = \int_0^\infty P_R\,dt = i_0^2 R\int_0^\infty e^{-2Rt/L}\,dt = \frac{i_0^2 RL}{-2R}\left(e^{-2Rt/L}\right)_0^\infty$$

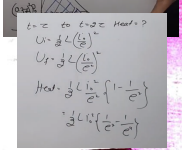
$$H_R = \frac{i_0^2 L}{-2}(0-1) = \frac{1}{2}Li_0^2$$

∴ Total heat dissipated will be equal to the energy stored in inductor at t = 0 and steady state change in magnetic

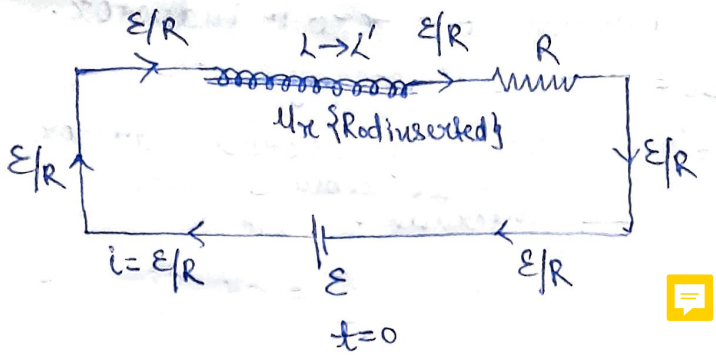
$$\boxed{H_R = \Delta U = U_i - U_f}$$







Q***



t=0

When current becomes ε/R, a rod is inserted $\{\mu_r\}$ in inductor and this situation is $t=0$. Find current in circuit as a function of time.

$L = \mu_0 n^2 \pi x^2 l$

$L' = \mu n^2 \pi x^2 l$

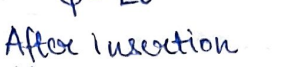
$= \mu_0 \mu_r n^2 \pi x^2 l$

$\therefore \boxed{L' = \mu_r L}$

Before insertion

$\phi = Li$

After insertion



$\phi = L' i'$

Now, we know flux will not change suddenly in inductor,

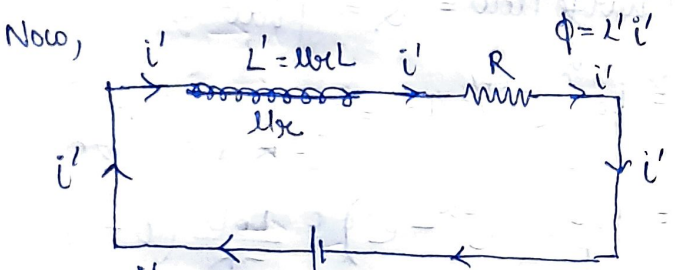
$\Rightarrow Li = L'i'$

$= Li = \mu_r L \cdot i'$

$\boxed{i' = \frac{i}{\mu_r}}$ current at $t=0$ after rod insertion

* We know that current also doesn't change suddenly but here as L is not fix, so current also changes to maintain same flux.

Now,



Using KVL: let at any time current be i

$\varepsilon - \mu_r L \frac{di}{dt} - iR = 0$

$= \varepsilon - iR = \mu_r L \frac{di}{dt}$

$= \int_0^t dt = \mu_r L \int_{\frac{\varepsilon}{\mu_r R}}^{i} \frac{di}{\varepsilon - iR}$

$= t = \frac{\mu_r L}{-R} \ln(\varepsilon - iR)_{\varepsilon/\mu_r R}^{i}$

$= \frac{-Rt}{\mu_r L} = \ln\left(\frac{\varepsilon - iR}{\varepsilon - \frac{\varepsilon}{\mu_r R}}\right)$

$= \frac{-Rt}{\mu_r L} = \ln\left(\frac{\varepsilon - iR}{\varepsilon - \frac{\varepsilon}{\mu_r}}\right)$

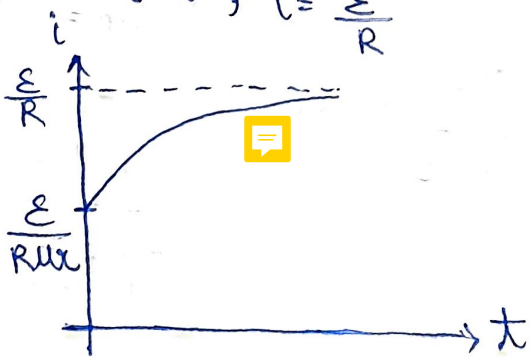
$= \frac{\varepsilon - iR}{\varepsilon\left(1 - \frac{1}{\mu_r}\right)} = e^{-Rt/\mu_r L}$

$= \varepsilon - iR = \varepsilon\left(1 - \frac{1}{\mu_r}\right) e^{-Rt/\mu_r L}$

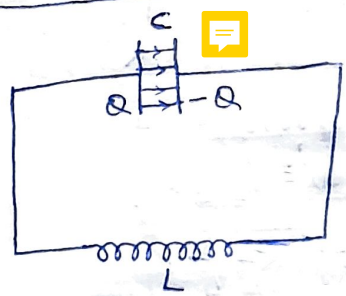
$\boxed{i = \frac{\varepsilon}{R}\left(1 - \left(1 - \frac{1}{\mu_r}\right) e^{-Rt/\mu_r L}\right)}$

$t=0, \; i = \frac{\varepsilon}{R\mu_r}$

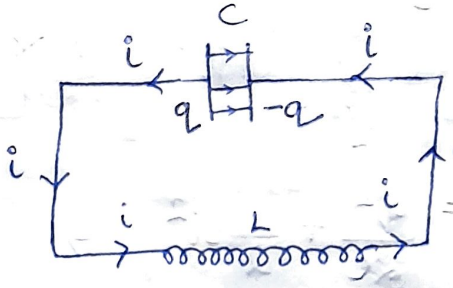
$t \to \infty, \; i = \frac{\varepsilon}{R}$



L-C Oscillations :



$t = 0$



$t = t$

Using KVL:

$$-\frac{(-q)}{C} - L\frac{di}{dt} = 0 \quad -(1)$$

$$i = -\frac{dq}{dt} \quad -(2)$$

From (1) and (2)

$$\frac{q}{LC} - \frac{d}{dt}\left(-\frac{dq}{dt}\right) = 0$$

$$\frac{q}{LC} + \frac{d^2q}{dt^2} = 0$$

$$\boxed{\frac{d^2q}{dt^2} + \frac{1}{LC}q = 0}$$

2nd Order D.E.

Now, Solu of D.E. $\equiv q = Q\sin\left(\frac{1}{\sqrt{LC}}t + \phi\right)$

$\{x = A\sin(\omega t + \phi)\}$

Now, at $t = 0$, $q = Q$

$Q = Q\sin(\phi) \Rightarrow \sin\phi = 1 \Rightarrow \boxed{\phi = \pi/2}$

$\therefore \star \boxed{q = Q\cos\left(\frac{1}{\sqrt{LC}}t\right)}$

$\boxed{\omega = \frac{1}{\sqrt{LC}}}$

$\boxed{T = \frac{2\pi}{\omega} = 2\pi\sqrt{LC}}$

$\boxed{f = \frac{1}{T} = \frac{1}{2\pi\sqrt{LC}}}$

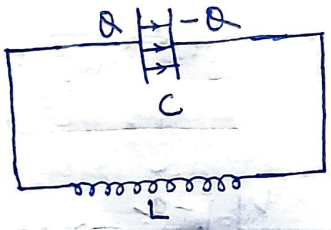
Now, $i = -\frac{dq}{dt}$

$$i = -\frac{Q}{\sqrt{LC}}\left(-\sin\left(\frac{1}{\sqrt{LC}}t\right)\right)$$

$\star \boxed{i = \frac{Q}{\sqrt{LC}}\sin\left(\frac{t}{\sqrt{LC}}\right)}$ $\star \boxed{i_{max} = \frac{Q}{\sqrt{LC}}}$

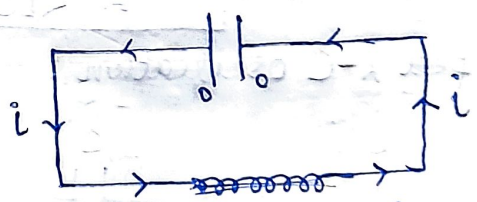
① $t = 0$, $q = Q$, $i = 0$

$t = T$



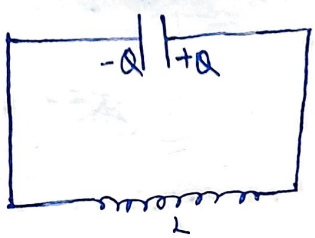
$U_e = \frac{Q^2}{2C}$, $U_m = 0$

② $t = \frac{T}{4} = \frac{\pi}{2}\sqrt{LC}$, $q = 0$, $i = \frac{Q}{\sqrt{LC}}$



$U_e = 0$, $U_m = \frac{1}{2}L\left(\frac{Q}{\sqrt{LC}}\right)^2 = \frac{Q^2}{2C}$

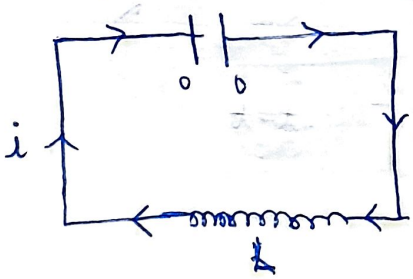
③ $t = T/2 = \pi\sqrt{LC}$, $q = -Q$, $i = 0$



$U_e = \frac{Q^2}{2C}$

$U_m = 0$

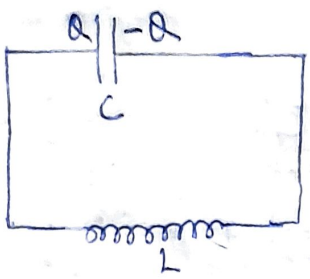
④ $t = \frac{3T}{4} = \frac{3\pi}{2}\sqrt{LC}$, $q = 0$, $i = -\frac{Q}{\sqrt{LC}}$

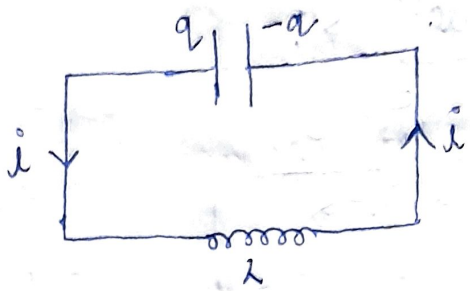


$U_e = 0$

$U_m = \frac{Q^2}{2C}$

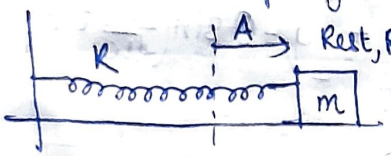
Note:





E/C $\quad \frac{Q^2}{2C} + 0 = \frac{q^2}{2C} + \frac{1}{2}Li^2 \rightarrow \boxed{i = \frac{1}{\sqrt{LC}}\sqrt{Q^2 - q^2}}$

Comparison from spring mass system:



$x \rightarrow q$

$v \rightarrow i$

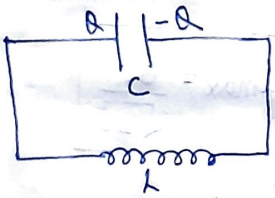
$\sqrt{\frac{k}{m}} \rightarrow \frac{1}{\sqrt{LC}}$

$k \rightarrow \frac{1}{C}$

$m \rightarrow L$

Spring mass	LC oscillation
$x = A\cos\omega t$	$q = Q\cos\omega t$
$\omega = \sqrt{\frac{k}{m}}$	$\omega = \frac{1}{\sqrt{LC}}$
$v = A\omega\sin\omega t$	$i = Q\omega\sin\omega t$
$= v_{max} = A\omega = A\sqrt{\frac{k}{m}}$	$i_{max} = Q\omega = \frac{Q}{\sqrt{LC}}$
$v = \omega\sqrt{A^2 - x^2}$	$i = \frac{1}{\sqrt{LC}}\sqrt{Q^2 - q^2}$

Q Find current in when capacitor has half of max^m charge.

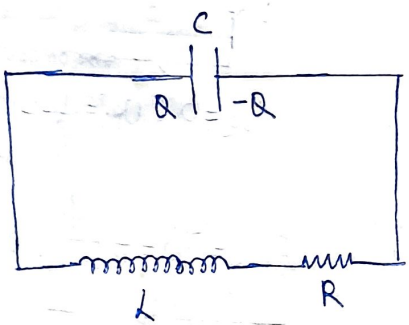


$E/C = \frac{Q^2}{2C} + 0 = \frac{(Q/2)^2}{2C} + \frac{1}{2}Li^2$

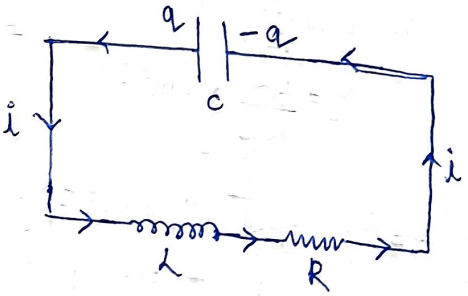
$= \frac{Q^2}{2C}\left\{1 - \frac{1}{4}\right\} = \frac{1}{2}Li^2 \Rightarrow \frac{3Q^2}{4LC} = i^2$

$\boxed{i = \frac{\sqrt{3}}{2}\frac{Q}{\sqrt{LC}}}$

Damped L-C oscillation:



$t = 0$



$t = t$

Using KVL: $-\left(-\frac{q}{C}\right) - L\frac{di}{dt} - iR = 0$

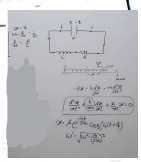
$= \frac{q}{C} - L\frac{di}{dt} - iR = 0$ $\quad i = -\frac{dq}{dt}$

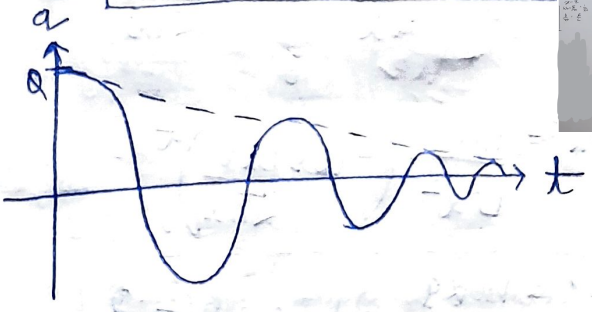
$= \frac{q}{C} + L\frac{d}{dt}\left(\frac{dq}{dt}\right) + R\frac{dq}{dt} = 0$

$= \boxed{\frac{d^2q}{dt^2} + \frac{R}{L}\frac{dq}{dt} + \frac{q}{LC} = 0}$

$\therefore \boxed{q = Qe^{-Rt/2L}\cos(\omega' t + \phi)}$

$\boxed{\omega' = \sqrt{\omega_0^2 - \left(\frac{R}{2L}\right)^2} = \sqrt{\frac{1}{LC} - \frac{R^2}{4L}}}$





Critical damping:

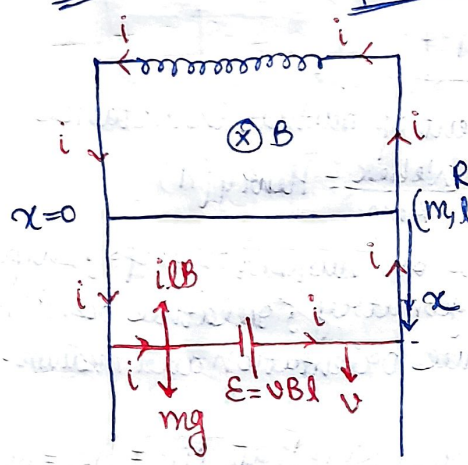
$\omega' = 0$

$\Rightarrow \frac{1}{LC} = \frac{R^2}{4L^2}$

$\star \boxed{R^2 = 4LC}$

in this case, the charge will decay exponentially

Q** Vertical plane fixed rail



At any time t, find ① velocity of rod ② position of rod ③ ~~current in circuit~~

① Using KVL:

$vBl - L\frac{di}{dt} = 0 \quad \text{—①}$

$mg - ilB = m\frac{dv}{dt} \quad \text{—②}$

Differentiating eqn ② wrt t.

$= -lB\frac{di}{dt} = m\frac{d^2v}{dt^2}$

Put di/dt in eqn ①

$vBl + L\left(\frac{m}{lB}\frac{d^2v}{dt^2}\right) = 0$

$\therefore \boxed{\frac{d^2v}{dt^2} + \frac{B^2l^2}{mL}v = 0}$

$v = v_0 \sin\left(\frac{Bl}{\sqrt{mL}}t + \phi\right)$

At $t=0$, $v=0$

$0 = v_0 \sin(\phi) \Rightarrow \boxed{\phi = 0}$

$v = v_0 \sin\left(\frac{Bl}{\sqrt{mL}}t\right)$

Now, $t=0$, $a = \frac{dv}{dt} = g = \frac{v_0 Bl}{\sqrt{mL}}\cos(0°)$

$\boxed{v_0 = \frac{\sqrt{mL}\,g}{Bl}}$

$\star \boxed{v = \frac{\sqrt{mL}\,g}{Bl}\sin\left(\frac{Bl}{\sqrt{mL}}t\right)}$

② Now,

$$\frac{dx}{dt} = v = \frac{\sqrt{mL}\,g}{Bl}\sin\left(\frac{Bl}{\sqrt{mL}}t\right)$$

$$\Rightarrow \int_0^x dx = \frac{\sqrt{mL}\,g}{Bl}\int_0^t \sin\left(\frac{Bl\,t}{\sqrt{mL}}\right)dt$$

$$= x = \frac{mLg}{B^2l^2}\left(-\cos\frac{Blt}{\sqrt{mL}}\right)_0^t$$

$$\boxed{x = \frac{mLg}{B^2l^2}\left\{1-\cos\frac{Blt}{\sqrt{mL}}\right\}}$$

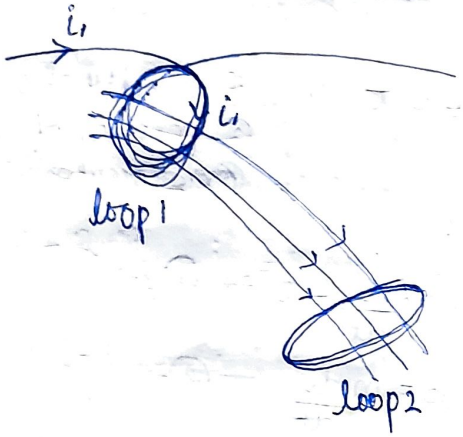
Also, at $t=0$, $x=0$

x_{max} is when $\cos\left(\frac{Blt}{\sqrt{mL}}\right) = -1$

$$\boxed{x_{max} = \frac{2mLg}{B^2l^2}}$$

$$\boxed{A = \frac{x_{max}}{2} = \frac{mLg}{B^2l^2}}$$

Mutual Inductance:



$\phi_2 \propto i_1$

$$\Rightarrow * \boxed{\phi_2 = M i_1}$$

M = Coefficient of mutual inductance

→ unit = $\frac{\text{Weber}}{\text{Amp}}$ = Henry (H)

→ depends on number of loops, shape, size, geometry, relative position, relative orientation and medium.

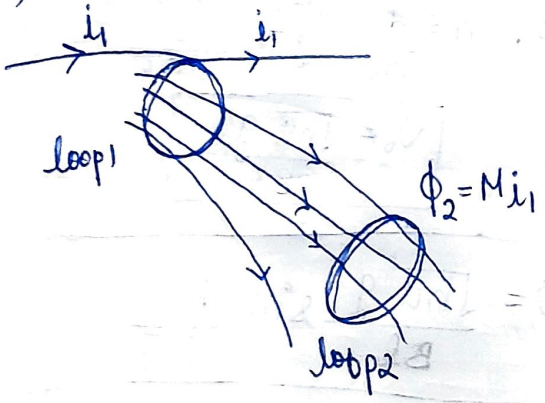
Reciprocity Theorem:

$\phi_1 \propto i_2$ $\quad$ $\phi_2 \propto i_1$

$\boxed{\phi_1 = M_{12} i_2}$ $\quad$ $\boxed{\phi_2 = M_{21} i_1}$

Now, this theorem says that $M_{12} = M_{21} = M$.
∴ For a pair of loops only one term of mutual inductance is defined.

Now, **alternate definition of mutual inductance** $\quad \boxed{\phi_2 = Mi_1}$ $\quad \boxed{\phi_1 = Mi_2}$



At $t = t$, i_1, $\phi_2 = Mi_1$

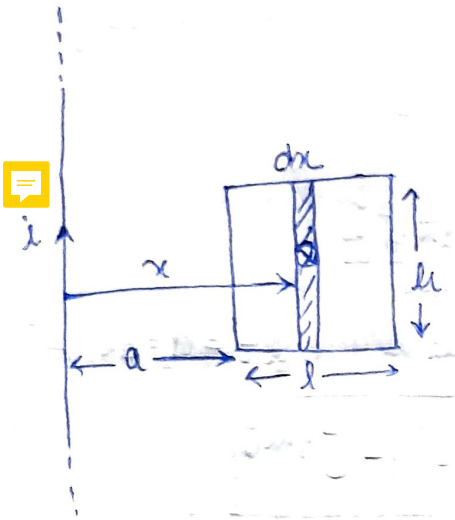
At $t = t+dt$, i_1+di_1, $\phi_2 + d\phi_2 = M(i_1+di_1)$

$d\phi_2 = M\,di_1$

$$\boxed{\mathcal{E}_2 = -\frac{d\phi_2}{dt} = -M\frac{di_1}{dt}}$$

↳ Emf induced in loop ② due to change in current in loop ①.

Q



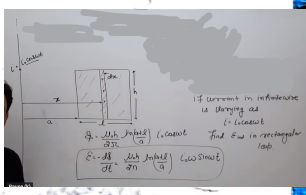
Find coefficient of mutual inductance

$$d\phi = \frac{\mu_0 i}{2\pi x} h\,dx \cos 0^\circ$$

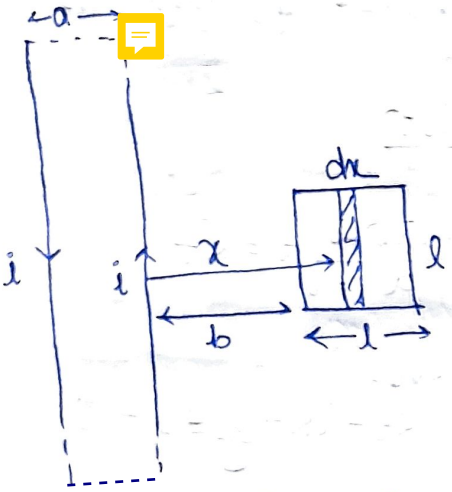
$$\phi = \int d\phi = \frac{\mu_0 i h}{2\pi} \int_a^{a+l} \frac{dx}{x} = \frac{\mu_0 i h}{2\pi} \ln\left(\frac{a+l}{a}\right)$$

$$\therefore \phi = \left\{\frac{\mu_0 h}{2\pi} \ln\left(\frac{a+l}{a}\right)\right\} i$$

$$\boxed{M = \frac{\mu_0 h}{2\pi} \ln\left(\frac{a+l}{a}\right)}$$



Q



Find coefficient of mutual inductance.

$$B = \frac{\mu_0 i}{2\pi x} - \frac{\mu_0 i}{2\pi (a+x)}$$

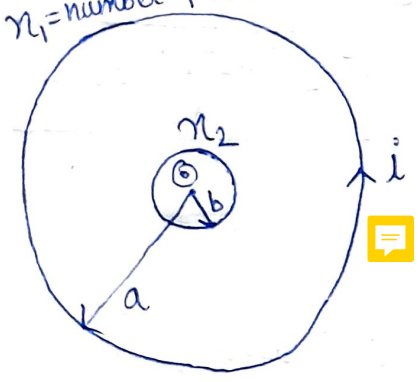
$$\phi = \int d\phi = \int_b^{b+l} \left(\frac{\mu_0 i}{2\pi x} - \frac{\mu_0 i}{2\pi (a+x)}\right) l\,dx$$

$$\phi = \frac{\mu_0 i l}{2\pi} \int_b^{b+l} \frac{dx}{x} - \frac{\mu_0 i l}{2\pi} \int_b^{b+l} \frac{dx}{a+x}$$

$$\phi = \frac{\mu_0 i l}{2\pi} \ln\left(\frac{b+l}{b}\right) - \frac{\mu_0 i l}{2\pi} \ln\left(\frac{a+b+l}{a+b}\right) = \left\{\frac{\mu_0 l}{2\pi} \ln\left[\frac{(b+l)(a+b)}{b(a+b+l)}\right]\right\} i$$

$$\therefore \boxed{M = \frac{\mu_0 l}{2\pi} \ln\left(\frac{(b+l)(a+b)}{b(a+b+l)}\right)}$$

Q



$b << a$

Concentric coplanar loops

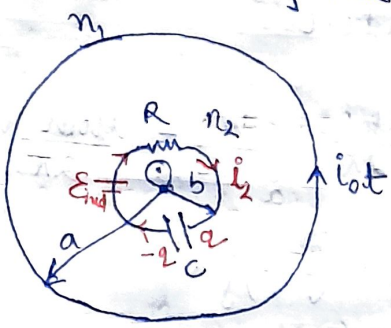
① Find M

$$B = \frac{\mu_0 n_1 i}{2a}$$

$$\phi = \frac{\mu_0 n_1 i}{2a} \cdot n_2 \pi b^2 \cos 0^\circ$$

$$\therefore \boxed{M = \frac{\mu_0 n_1 n_2 \pi b^2}{2a}}$$

(2) If current in outer loop is varying as $i = i_0 t^2$, then find charge on capacitor at any time t.



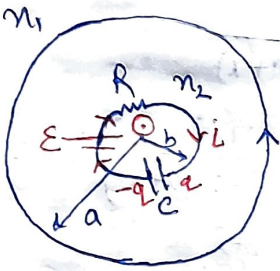
$$\phi = \frac{\mu_0 n_1 n_2 \pi b^2}{2a} i_0 t$$

$$\mathcal{E} = \frac{d\phi}{dt} = \frac{\mu_0 n_1 n_2 \pi b^2 i_0}{2a}$$ = constant; thus same as charging RC circuit

Charge on capacitor at any time $t = q = \mathcal{E}C\left(1 - e^{-t/RC}\right)$

$$\boxed{q = \frac{\mu_0 n_1 n_2 \pi b^2 i_0 C}{2a}\left\{1 - e^{-t/RC}\right\}}$$

(3) If current in outer loop is varying as $i = i_0 t^2$, then find current in smaller loop at any time t.



$$\phi = \frac{\mu_0 n_1 n_2 \pi b^2 i_0 t^2}{2a}$$

$$\mathcal{E} = \frac{d\phi}{dt} = \left(\frac{\mu_0 n_1 n_2 \pi b^2 i_0}{a}\right) t = kt$$

$$i = \frac{dq}{dt} \quad (1)$$

Using KVL:

$$kt - iR - \frac{q}{C} = 0 \quad (2)$$

Differentiate (2) wrt time,

$$k - R\frac{di}{dt} - \frac{i}{C} = 0$$

$$= \frac{kC - i}{CR} = \frac{di}{dt}$$

$$= \int_0^t \frac{dt}{RC} = \int_0^i \frac{di}{kC - i}$$

$$= \frac{t}{RC} = -\ln(kC - i)_0^i$$

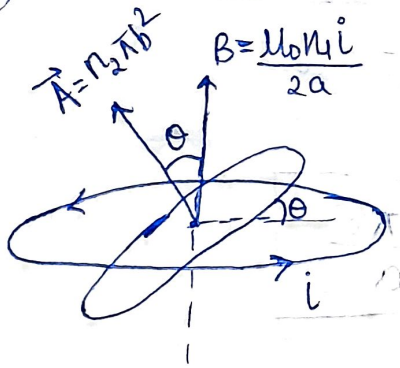
$$= -\frac{t}{RC} = \ln\left(\frac{kC - i}{kC}\right)$$

$$= e^{-t/RC} = 1 - \frac{i}{kC}$$

$$\boxed{i = kC\left\{1 - e^{-t/RC}\right\}}$$

$$\boxed{i = \frac{\mu_0 n_1 n_2 \pi b^2 i_0 C}{a}\left\{1 - e^{-t/RC}\right\}}$$

(4)



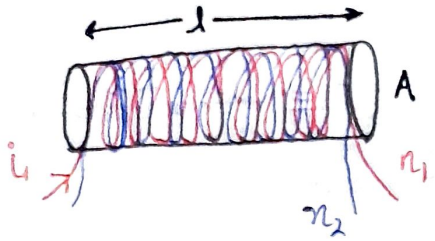
Find M

$$\phi = \frac{\mu_0 n_1 i}{2a} n_2 \pi b^2 \cos\theta$$

$$\phi = \left\{\frac{\mu_0 n_1 n_2 \pi b^2 \cos\theta}{2a}\right\} i$$

$$\boxed{M = \frac{\mu_0 n_1 n_2 \pi b^2 \cos\theta}{2a}}$$

Relation between mutual inductance and self inductance :



$L_1 = \mu_0 n_1^2 A l$

$L_2 = \mu_0 n_2^2 A l$

$M = \mu_0 n_1 n_2 A l = \sqrt{\mu_0 n_1^2 A l}\,\sqrt{\mu_0 n_2^2 A l}$

$\therefore \star \boxed{M = \sqrt{L_1 L_2}}$

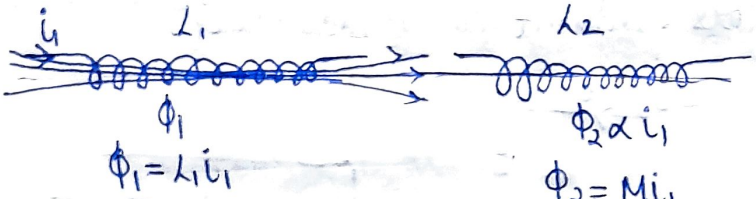
- only if flux is completely linked
- No leakage of flux
- MLOF passing through loop 1 is also passing through loop 2.

$\phi_2 = \mu_0 n_1 i_1 A n_2 l$

$\phi_2 = (\mu_0 n_1 n_2 A l) i_1$

$\therefore M = \mu_0 n_1 n_2 A l$

But, in general



$\phi_1 = L_1 i_1$

$\phi_2 = M i_1$

$\phi_2 < \phi_1 \Rightarrow M < \sqrt{L_1 L_2}$

↳ Exact value depends on relative position, orientation, and medium.

So, $\boxed{M = K\sqrt{L_1 L_2}}$

K = Coupling Constant, $\boxed{K \le 1}$

① If $K = 1 \Rightarrow$ No leakage.

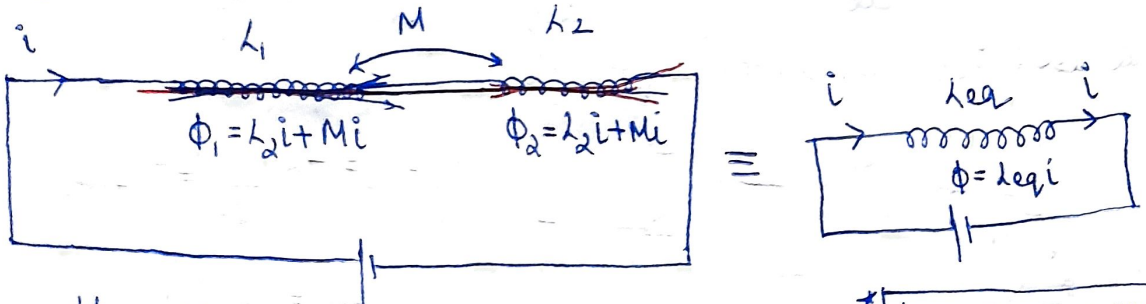
② near $\quad K \approx 1$

③ Very far $\quad K \approx 0$

④ $K = 0$

Inductor in Series :

① Aiding Series Combination :



Potential difference $= -\frac{d\phi_1}{dt} - \frac{d\phi_2}{dt} = -\frac{d}{dt}(L_1 i + Mi) - \frac{d}{dt}(L_2 i + Mi)$

$\star (L_1 + L_2 + 2M)\frac{di}{dt} = -\frac{d}{dt} L_{eq} i = \star L_{eq}\frac{di}{dt}$

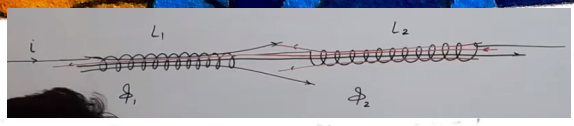
$\therefore \star \boxed{L_{eq} = L_1 + L_2 + 2M}$

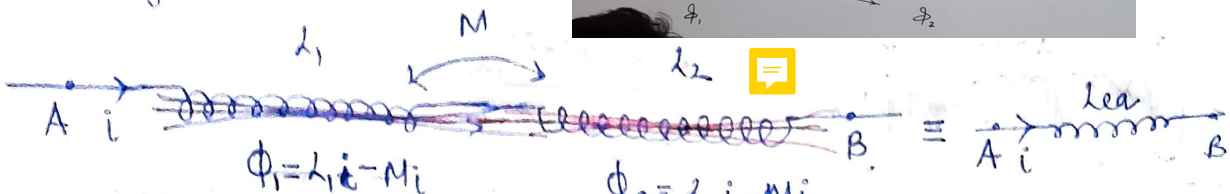
$\boxed{L_{eq} = L_1 + L_2 + 2K\sqrt{L_1 L_2}}$

If large sep ⇒ decoupled

$\star \boxed{L_{eq} = L_1 + L_2}$

(2) Opposing series combination :





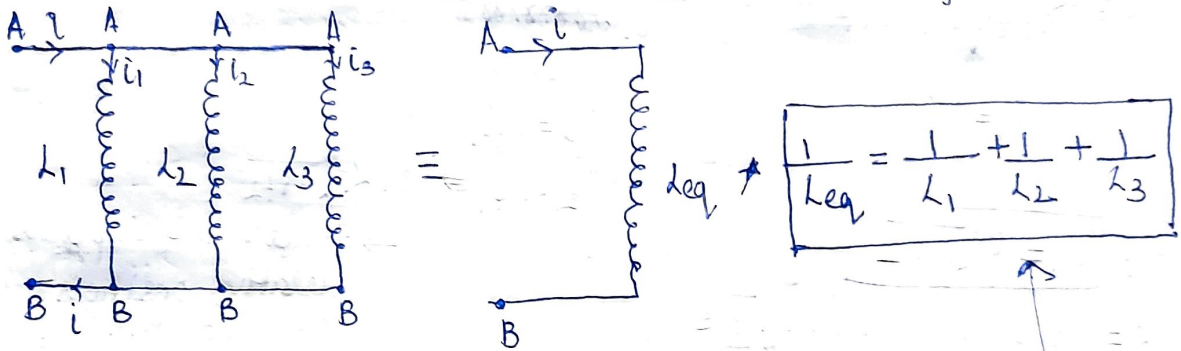
$\phi_1 = L_1 i - Mi$ $\qquad$ $\phi_2 = L_2 i - Mi$

$$V_A - V_B = -\frac{d\phi_1}{dt} - \frac{d\phi_2}{dt} = -\frac{d}{dt}\{L_1 i - Mi\} - \frac{d}{dt}\{L_2 i - Mi\}$$

$$= -(L_1 + L_2 - 2M)\frac{di}{dt} = -L_{eq}\frac{di}{dt}$$

$$\therefore \boxed{L_{eq} = L_1 + L_2 - 2M} \qquad \boxed{L_{eq} = L_1 + L_2 - 2k\sqrt{L_1 L_2}}$$

Parallel Combination of inductors {without mutual inductance}:



$$\boxed{\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3}}$$

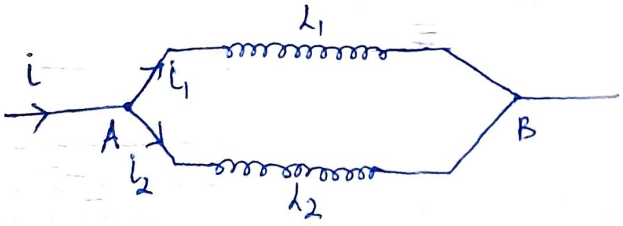
$$V_A - V_B = -L_1\frac{di_1}{dt} \qquad V_A - V_B = -L_{eq}\frac{di}{dt}$$

$$V_A - V_B = -L_2\frac{di_2}{dt} \qquad \text{Now, } i = i_1 + i_2 + i_3$$

$$V_A - V_B = -L_3\frac{di_3}{dt} \qquad = \frac{di}{dt} = \frac{di_1}{dt} + \frac{di_2}{dt} + \frac{di_3}{dt}$$

$$= \frac{V_A - V_B}{L_{eq}} = \frac{V_A - V_B}{L_1} + \frac{V_A - V_B}{L_2} + \frac{V_A - V_B}{L_3}$$

Special Case : For two inductors in parallel.



$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2}$$

$$\boxed{L_{eq} = \frac{L_1 L_2}{L_1 + L_2}}$$

$$V_A - V_B = -L_1\frac{di_1}{dt} = -L_2\frac{di_2}{dt}$$

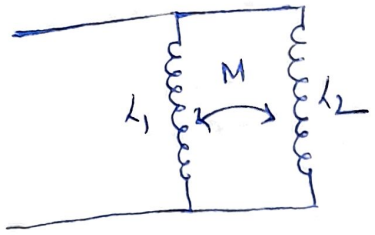
$$= L_1\int_0^{i_1} di_1 = L_2\int_0^{i_2} di_2$$

$$\boxed{L_1 i_1 = L_2 i_2}$$

$$\boxed{\frac{i_1}{i_2} = \frac{L_2}{L_1}}$$

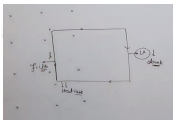
Parallel Combination with Mutual inductance :

① Aiding Parallel combination ⇒ $L_{eq} = \dfrac{L_1L_2 - M^2}{L_1 + L_2 - 2M}$

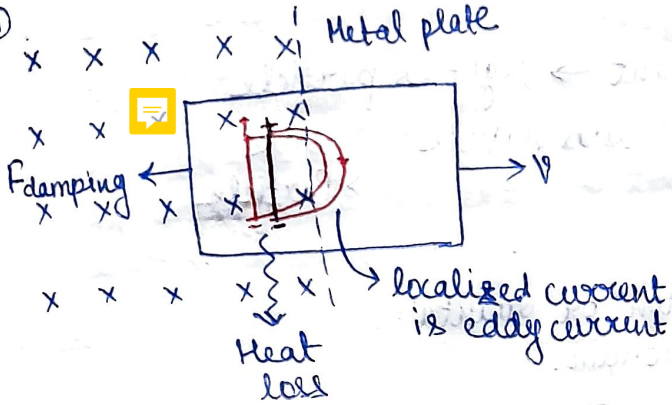


② Opposing Parallel combination ⇒ $L_{eq} = \dfrac{L_1L_2 - M^2}{L_1 + L_2 + 2M}$

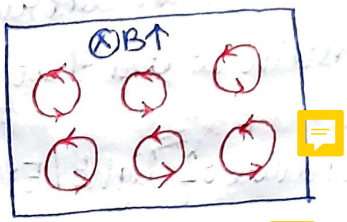
Eddy Current :



① Metal plate

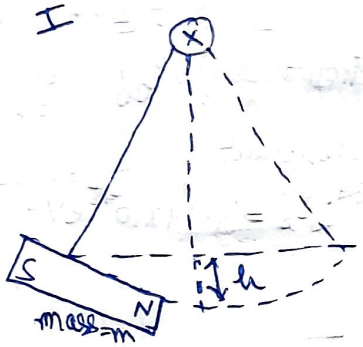


② Metal plate

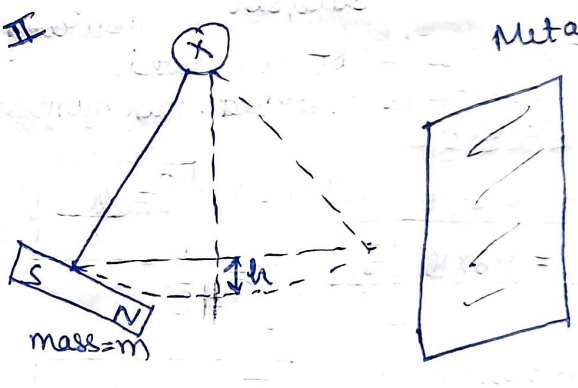


→ Localised Current in the metal; due to relative motion b/w metal and magnetic field or due to TVMF is known as eddy current.

Q I

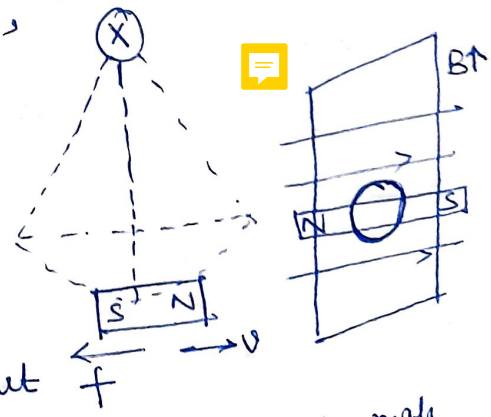


II



which will stop first?

In case I, there is no damping, so it will never stop. But is case II, when magnet approaches the sheet, magnetic field passing through sheet increases, due to this an eddy current is induced in such a way so as to decrease the increasing magnetic field. The force applied by the sheet has -ve work done and hence releases energy of the system. When the magnet is going back, opposite will happen but the work done will still be -ve and this will provide damping and magnet will stop after some time.



Heat = mgh

The potential energy that magnet had initially gets lost as heat due to the eddy currents